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Digitalisation, Data, Emerging Innovations, Exchange of Data

SFERA Protocol: Train Operation Data Exchange with On-Board Devices



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All Railway Undertakings and Infrastructure Managers adopting Driver Advisory Systems (DAS)

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Driver Advisory Systems; DAS; Standalone DAS; S-DAS; Connected DAS; C-DAS; SFERA; data exchange; Traffic Management System; TMS; ATO over ETCS; EoA; SUBSET-126; ETCS; ERTMS; punctuality; energy efficiency; energy savings; cost savings; interoperability; timetables; capacity

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Record of updates:

1st edition, June 2020	Creation of the document
2nd edition, August 2022	2nd edition of this IRS

1. Integration of SUBSET-126 updates
Reason: SFERA data is designed to be a superset of SUBSET-126

2. Introduction of "timetable mode"
Reason: Request from WG members. It is believed that timetable mode eases the transition from paper information to DAS.

3. Addition of real-time information in Status Report (e.g. measured current/voltage, battery level).
Reason: Request from WG members. Provides additional information from board to ground.

4. Introduction of Advised Speed in Temporary Constraints.
Reason: Request from WG members. Gives the option in C-DAS-O to provide an explicit advised speed when needed.

5. *Use of TAF/TAP Common Interface (CI) for IM-RU communication.*
Reason: Request from WG members. Eases the adoption of SFERA as CI is already implemented.

6. Review of DAS Handshake.
Reason: The handshake process in Edition 1 needed improvement in consistency and explanations.

7. Addition of Network-Specific Parameter in the XSD.
Reason: Improves flexibility in the data structure for local customisations or tests. May be used for network-specific needs or to test a new feature before asking for a Change Request.

8. Reverting to SUBSET-126 Stopping Point logic. Each Timing Point may include at most one Stopping Point, but several Stopping Points may be grouped in a Stopping Zone.
Reason: Improves compatibility with SUBSET-126.

9. Addition of information on occupation and status of other trains (in front or behind).
Reason: Request from WG members, to adapt the data to their already existing tools. Provides more comprehensive information to the driver.

10. Introduction of TAF/TAP locations.
Reason: Request from WG members, allowing to make the link between SFERA messages and TAF/TAP messages.

3rd edition, January 2026	3rd edition of this IRS

1. Extension of SFERA from DAS-only to a multi-function operational protocol
Reason: To evolve SFERA from a Driver Advisory System interface into a general train-operation data exchange protocol supporting driver advisory messages, digital instructions and free text communication use cases.

2. *Introduction of Digital Instructions (DI) as a SFERA function*
Reason: To support the digital exchange and lifecycle management of operational instructions in line with TSI OPE requirements.

3. *Introduction of Free Text Communication (FTC) as a SFERA function*
Reason: To enable structured, traceable, multilingual non-safety-critical text communication between ground systems and drivers using the SFERA framework.

4. *Extension and clarification of use cases (Appendix A)*
Reason: To include the DI and FTC functions in the SFERA use cases.

5. *Introduction of function-based separation in topic tree (Appendix B)*
Reason: To allow different operational functions (DAS, DI, FTC) to coexist on the same MQTT infrastructure while remaining logically separated.

6. *Addition of DI and FTC message payload families (B2G/G2B Event/Request/Reply) (Appendices D, E)*
Reason: The SFERA XSD version 4.00 expands SFERA beyond DAS by introducing dedicated payload structures for Digital Instructions (DI) and Free Text Communication (FTC), so the same protocol can carry those functions' data.

7. *Introduction of a sample XSLT-based processor for European Instructions (new Appendix I)*
Reason: To provide a concrete, implementable example of how SFERA Digital Instruction XML can be transformed into human-readable form on board, supporting multilingual rendering, layout customisation and reuse of a common DI data model.

8. *Creation of a centralised set of supporting and reference documents (new Appendix J)*
Reason: To improve transparency, maintainability and evolution of SFERA by hosting reference documents (e.g. NSP lists, implementation parameters, protocol mappings) in a public repository, decoupled from the normative IRS text.

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Foreword

An International Railway Solution (IRS) is a document drawn up by consensus and applied on a voluntary basis which aims to facilitate and harmonise railway operation.

IRSs are gradually replacing UIC leaflets.

Members of the International Union of Railways (UIC) participate in the creation of IRSs, of which they benefit in full. Furthermore, marketing to railway stakeholders contributes to the sharing and harmonisation of good practices, in accordance with UIC's primary goal.

While IRSs do not have the force of law, they are created and published in a professional manner and with technical approval on the basis of UIC procedures and have full technical legitimacy in the railway sector.

The "General application" section constitutes the heart of the IRS and contains the working group's expertise applicable in the broadest manner. The addition of a "Specific application" aims to cover factors specific to a region, a type of service, etc.

Some provisions shall be implemented in all cases to ensure correct application and maintain the system's cohesiveness. Certain operational circumstances require that specific provisions be systematically applied by all users of the document.

For the purposes of understanding, the user acknowledges that the expression of:

- obligation indicates requirements to be respected in order to comply with the expertise expressed in the document (usually "shall");

- recommendation indicates options that are favourable but not essential (usually "should");

- permission indicates options that are recognised but without consequence (usually "may");

- capacity indicates possibilities from a technical standpoint and not permissions or probabilities (usually "can").

Obligation, recommendation, and capacity can be expressed in the negative to indicate prohibiting, non-recommendation, and incapacity.

If applicable, notably for the purposes of compliance with a UIC approval, passages that are recommended or mandatory for the cohesiveness of the system will be accompanied by the letters "R" and "MCS" respectively.

This IRS reflects the following UIC working group expertise:

UIC SFERA (Smart communication For Efficient Railway Activity) Project Working Group

It integrates in full or in part the following UIC leaflets:

- none

It complements the following standards:

- none

The following IRSs constitute a complementary series:

- none

Executive summary

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

This IRS is the result of collaborative work carried out by leading railway stakeholders between 2017 and 2025 under the auspices of the UIC Sustainability Platform and the Rail System Forum. External stakeholders also contributed through workshops and draft document sharing.

The SFERA protocol serves as a messaging standard for train operation data exchange between infrastructure managers (IMs), railway undertakings (RUs), and on-board devices.

Driver Advisory Systems (DAS) are IT tools which provide drivers with information to help them arrive at stations on time[1] while reducing energy consumption. DAS take into account various types of data, such as live position tracking, infrastructure characteristics, real-time transport plans, rolling stock data, and so on. They can use the SFERA protocol to accommodate Automatic Train Operations (ATO) as well as DAS applications, working both over the European Train Control System (ETCS) and conventional Automatic Train Protection systems (also referred to as Class B ATP).

Digital Instruction (DI) systems are specialised to create, digitally transmit and manage the lifecycle of instructions as prescribed by *TSI OPE*, facilitating communication between signallers and drivers. Additionally, the SFERA protocol enables the cross-border application of DI for specific solutions implemented by IMs or RUs.

Free Text Communication (FTC) enables the exchange of text messages between the driver and ground-based systems and personnel, with a primary focus on IM stakeholders. The role of this function is to handle information exchange that is not related to:

- Optimising traffic flow (handled via the DAS function)

- Transmitting operational instructions, which are safety-related communication and shall be communicated through the DI function

It integrates in full or in part the following UIC leaflets:

- none

It complements the following standards:

- none

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General application

Chapter 1 gives a general introduction into DAS, DI and FTC terminology, the context of their implementation, the purpose for designing a harmonised method of handling data exchange between trackside and on-board systems, and in line with the ATO implementation strategy.

1. For example, a punctuality analysis by TTG in Switzerland in 2017 showed an average improvement of 19% in punctuality (without DAS: 42% on-time (0s) with a maximum ± 30s window / with DAS: 50% on-time (0s) with a maximum ± 30s window).

Chapter 2 sets out a list of requirements for the implementation of the SFERA protocol, relating to the expected business outcomes

Chapter 3 defines the different functions of the exchange protocol, which have been defined as DAS, DI, and FTC.

Chapter 4 defines use cases which describe the main types of interaction between on-board SFERA functions and ground systems or external entities, as well as the implementation prerequisites to be agreed upon between RU and IM.

Chapter 5 gives an overview of the structure of the SFERA data messaging system.

Chapter 6 deals with communication mechanisms, the protocol layers, and design specifications.

Chapter 7 defines the relevant maintenance needs and procedures to keep the SFERA protocol up to date.

Appendices

This IRS also proposes a series of appendices containing guidelines for implementing SFERA to avoid the most foreseeable pitfalls.

Appendix A provides complete details of the SFERA use cases touched on in Chapter 4.

Appendix B provides an in-depth description of the SFERA Communication Protocol defined in Chapter 6.

Appendix C covers implementation cases, i.e. selected real-life scenarios that illustrate the sequence of use cases and the communication-level events occurring within a specific setup.

Appendix D contains the SFERA XML Schema Definition, i.e. the content and format of the data that can be exchanged in SFERA.

Appendix E is a handbook which gives detailed information on the SFERA data structure as well as guidelines and implementation suggestions.

Appendix F provides examples of XML files compatible with the SFERA format.

Appendix G contains a correspondence table between SUBSET-126 and the SFERA elements.

Appendix H is a reference table for error codes in SFERA.

Appendix I contains sample XSLT files that can be used as the basis for a workable solution to transform SFERA XML messages into HTML documents.

Appendix J lists supporting documents that are made available by the UIC SFERA Working group, for complementary information outside the IRS itself.

Field of application

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

This IRS applies to the digital data exchange for train operations between railway systems of infrastructure managers and railway undertakings, focusing on communication with on-board systems. It defines a standard that is referred to as the Smart communications For Efficient Rail Activities (SFERA) Protocol.

The protocol supports the implementation of the following SFERA functions:

- Driver Advisory Systems (DAS), whether connected or standalone,
- Digital Instructions (DI), as operational instructions between the signaller and train driver,
- Free Text Communication (FTC), for contextual or situational messaging between stakeholders.

The SFERA protocol provides the option to implement these functions independently or in combination, and, where applicable, be integrated into Traffic Management Systems (TMS) or Automatic Train Operation (ATO) systems.

The SFERA protocol includes:

- The content and format of the data exchanged between systems,
- The use cases relevant to data exchange involving the different functions,
- The communication architectures,
- The basic functional requirements necessary to ensure system compatibility.

The SFERA protocol does not specify:

- Safety requirements or processes related to safety
  Note: The safety implications of introducing a SFERA function should be assessed in each specific implementation case. Particular attention should be paid to the processes generating, (re)calculating, or presenting data, whether used for data exchange or for the user interface.
- The design, behaviour, or presentation of user interfaces
- The calculation logic for messages such as driver advice
- The implementation of a specific solution
  Note: This document only describes the interface. While certain use cases are defined, working arrangements between railway undertakings and infrastructure managers may differ between countries. Responsibilities should therefore be divided accordingly as part of a specific national implementation.

- The implementation scope of a SFERA function
  Note: This IRS is designed to support both basic and advanced implementations of DAS, DI, FTC and, where applicable, TMS, enabling railway undertakings and infrastructure managers to define flexible migration processes.

- Implementation-specific values and data that may be commercially sensitive and are unique to each implementation.

For these aspects, a range of requirements and best practices already exist across different implementations of systems such as DAS and ATO. While these factors are essential for overall system performance, this document focuses solely on the requirements related to data exchange.

This document supports a harmonised implementation of the protocol for:

- Railway undertakings and infrastructure managers, whether currently using or planning to introduce one or more SFERA functions,

- System and solution suppliers, to facilitate the development and implementation of products aligned with the SFERA protocol,

- IT system and software providers offering related services, to support the specification and implementation of system interfaces.

Reference documents

The following documents are cited in this IRS or strongly support its requirements and recommendations:

1. European Standards

International Union of Railways (UIC) - Railway Interchange Coding System (RICS)

[1] *UIC RICS Codes,* December 2024. Available: https://uic.org/rics

European Committee for Standardisation (CEN)

[2] *EN 13816: Transportation - Logistics and services - Public passenger transport - Service quality definition, targeting and measurement,* September 2002

[3] *EN 61209: Maritime navigation and radio communication equipment and systems. Integrated bridge systems (IBS). Operational and performance requirements, methods of testing and required test results,* November 1999

Note: EN standards can be purchased through National Standardisation Bodies.

European Union - Agency for Railways (ERA)

[4] *ERA: "List of Class B Systems": ERA/TD/2011-11 v4.0.,* Available: https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf.

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European Rail Traffic Management Systems (ERTMS)

[6] *ETCS Variables.* Available: https://www.era.europa.eu/content/etcs-variables

European Union Directives and Regulations

[7] *Directive (EU) 2012/34 of the European Parliament and of the Council establishing a single European railway area,* November 2012. Available: https://eur-lex.europa.eu/homepage.html

[8] *Directive (EU) 2016/797 of the European Parliament and of the council of 11 May 2016 on the interoperability of the rail system with the European Union,* May 2016. Available: https://eur-lex.europa.eu/homepage.html

[9] *Directive 2009/72/EC of the European Parliament and of The Council concerning common rules for the internal market in electricity*, July 2009. Available: https://eur-lex.europa.eu/homepage.htm

[10] *COMMISSION IMPLEMENTING REGULATION (EU) 2019/773 of 16 May 2019 on the technical specification for interoperability relating to the operation and traffic management subsystem of the rail system within the European Union and repealing Decision 2012/757/ EU,* May 2019 (Appendices A, C2, J), amended by (EU) 2020/778, (EU) 2021/2238, (EU) 2023/1693, and (EU) 2025/458. Available: https://eur-lex.europa.eu/homepage.htm

Automatic Train Operation (ATO)

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Internet Engineering Task Force (IETF) - Request for Comment (RFC)

[12] *A Universally Unique IDentifier (UUID) URN Namespace, IETF RFC 4122.* Available: https://tools.ietf.org/html/rfc4122

[13] *The WebSocket Protocol, IETF RFC 6455.* Available: https://tools.ietf.org/html/rfc6455

[14] *JSON web token standard, IETF RFC 7519.* Available: https://tools.ietf.org/html/rfc7519

[15] *The Transport Layer Security (TLS) Protocol Version 1.3, IETF RFC 8466.* Available: https://tools.ietf.org/html/rfc8446

Technical Specifications for Interoperability (TSI) - Telematic Applications Passengers (TAP) and Telematic Applications Freight (TAF)

[16] *TAP TSI and TAF TSI Sector Handbook for the Communication between Railway Undertakings and Infrastructure Managers (RU/IM Telematics Sector Handbook).* Available: http://taf-jsg.info/

[17] *TAF TSI - ANNEX D.2: APPENDIX E - COMMON INTERFACE.* Available: https://github.com/EU-Agency-for-Railways/TSI_TAF

[18] *TAF TSI - ANNEX D.2: APPENDIX F - TAF TSI DATA AND MESSAGE MODEL* . Available: https://github.com/EU-Agency-for-Railways/TSI_TAF

European Union Funding for Research & Innovation

[19] *ON-TIME project Deliverable 6.1: Specifications of a driving advisory systems (DAS) data format.* Available: https://cordis.europa.eu/project/id/285243/reporting.

Book in German

[20] *"Elektrische Bahnen"*, Author: Žarko Filipović, publisher: Springer-Verlag, 2015.

The Organization for the Promotion of Structured Information Standards (OASIS) - Message Queuing Telemetry Transport (MQTT)

[21] *MQTT Version 5.0 - OASIS Standard,* March 2019. Available: https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.pdf

World Wide Web Consortium (W3C)

[22] *XSL Transformations (XSLT) Version 3.0, W3C Recommendation*, June 2017. Available: https://www.w3.org/TR/xslt-30/

Abbreviations and Symbols

ATO	Automatic Train Operations
ATP	Automatic Train Protection
B2G	Board to Ground
C-DAS	Connected DAS
C-DAS-C	Connected DAS with Centralised intelligence
C-DAS-O	Connected DAS with On-board intelligence
CI	Common Interface
DAS	Driver Advisory System
DI	Digital Instruction
DMI	Driver machine interface
EI	European Instruction
EoA	End of Authority
ERTMS	European Rail Traffic Management System
ETCS	European Train Control System
G2B	Ground to Board
GNSS	Global Navigation Satellite System
GoA	Grade of Automation
IM	Infrastructure Manager
IRS	International Railway Solution
JP	Journey Profile
JSON	JavaScript Object Notation
MQTT	Message Queuing Telemetry Transport
NI	National Instruction
NTC	[ETCS Level] National Train Control

OBU	On-Board unit
OPE	Operation and traffic management
PLC	Primary Location Code
RU	Railway Undertaking
S-DAS	Standalone DAS
SFERA	Smart communications For Efficient Rail Activities
SFERA-OB	SFERA On-Board
SFERA-OB/xx	SFERA On-Board for xx-function (e.g. DAS, DI or FTC)
SFERA-TS	SFERA Trackside
SFERA-TS/xx	SFERA Trackside for xx-function (e.g. DAS, DI or FTC)
SLC	Subsidiary Location Code
SP	Segment Profile
S2R	Shift2Rail
SR	Status Report
TAF	Telematic Applications Freight
TAP	Telematic Applications Passengers
TC	Train Characteristics
TCMS	Train Control and Management System
TCP/IP	Transmission Control Protocol (TCP) over the Internet Protocol (IP)
TIU	Train Interface Unit
TMS	Traffic management system
TP	Timing Point
TPE	Train path envelope
TSI	Technical Specifications for Interoperability
TSO	Transmission system operator
TSR	Temporary speed restriction

TTCP	TimeTable control point
UTC	Coordinated Universal Time
VKM	Vehicle Keeper Marking
XML	eXtensible Markup Language
XSD	XML Schema Definition
XSLT	eXtensible Stylesheet Language Transformations

Concepts, terms and definitions

(See Reference documents - page 7)

1. Access code

A unique alphanumeric code (letters and/or digits) used to authenticate a receiver as the correct recipient and to establish the ability to communicate or interact regarding an instruction.

Note 1: The access code enables a mechanism for two-way verification and authorisation, ensuring authenticated interactions with the system concerning an instruction.

2. Authentication

A security process that verifies the legitimacy of a transmission, message, or originator by confirming the identity of a user, process, or device, typically as a prerequisite for granting access to resources within an information system.

Note 1: Authentication is a prerequisite for access to resources in a control system. It ensures that a claimed identity characteristic is accurate and trustworthy.

3. Authorisation

The act of granting someone official permission to perform a specific action.

4. Business tool

A software application or interface, with or without direct user interaction, that performs operational or communication functions and acts as the source or recipient of structured data exchanged in a railway business process.

Note 1: A business tool may be actively used by a human operator (e.g., driver HMI, mobile app) or may operate autonomously (e.g., Traffic Management System generating driver advice).

Note 2: It represents the end system or component where business-level functions are executed and where structured information objects are created or consumed.

5. Common Interface

A module defined in the TAF/TAP TSI and deployed at the boundary of a railway stakeholder's IT systems.

Note 1: It supports the exchange of data between internal systems (which typically use non-interoperable protocols) and external systems, using an interoperable protocol.

Note 2: It may also be used to transmit specific messages that are not covered by the standard messages defined in the *TAF/TAP TSI* and the sector handbook.

6. Digital Instruction (DI)

An operational instruction communicated by digital means, whether supported by other forms of communication or not.

7. Driving profile

Set of advice presented to the driver during the journey to optimise train operation, based on real-time data, the train path envelope, and all other applicable constraints.

Note 1: The advice is computed on board by an algorithm.

Note 2: Examples of advice include: “traction”, “cruise”, “coast” and “brake”.

8. European Instruction (EI)

A harmonised operational instruction giving similar content to train drivers across the European Union in order for them to answer in a similar manner to similar situation.

[*2019/773/EU - TSI OPE - Appendix J - Glossary*]

9. Grade of Automation (GoA)

The level of automation at which a train can operate, resulting from basic train operation function responsibilities being shared between operations staff and systems.

Note 1: The Grades of automation are:

- GoA1: Non-automated train operation. The train is manually driven, but protected by automatic train protection (ATP), including trains with DAS,

- GoA2: Semi-automated train operation. ATO is used with a train driver present in the cab.

- GoA3: Driverless train operation. A train attendant is nevertheless on board.

- GoA4: Unattended train operation.

Note 2: The definition is derived from *Subset-125 of ERTMS/ATO*.

10. Handshake

The process of pairing the intended sender and receiver of SFERA messages establishing a communication link between the two parties.

11. interoperability

The ability, including in the light of all the regulatory, technical, administrative and operational conditions, of the infrastructure, including digital infrastructure in a transport mode or segment and between different transport modes, to allow safe and uninterrupted traffic and information flows which achieve the required levels of performance for that infrastructure mode or segment.

[*2024/1679/EU - Art.3*]

12. Instruction lifecycle

The progression of a Digital Instruction through all phases, from creation to cancellation, revocation, or completion, where certain phases may require the prior completion of specific activities or the attainment of certain statuses.

13. Journey

The scheduled movement of a vehicle along a single route.

Note 1: The definition is derived from *EN 13816:2002*.

14. Journey Profile (JP)

A dataset comprising the list of timing points, references to Segment Profiles and Train Characteristics, and temporary changes to the infrastructure (e.g. TSRs).

Note 1: The Journey Profile is defined in real time based of the train path envelope. It is sent from ground to board systems to calculate the driving profile.

15. Location

The location of an object or correspondent, which may be expressed in different formats.

16. Message

A package of data transmitted between TS and OB systems, in either direction.

Note 1: A message includes both the payload and the metadata required for processing and delivery.

17. National instruction (NI)

An instruction defined at national level or by an infrastructure manager which covers situations specific to a Class B system or the transition between Class A and Class B systems.

[*2019/773/EU - TSI OPE - Appendix J - Glossary*]

18. Operating language

A language or languages used in daily operation by an IM and published in its Network Statement, for the communication of operational or safety related messages between the staff of the IM and RU.

[*2019/773/EU - TSI OPE - Appendix J - Glossary*]

19. Operational instruction

Formal information exchanged between signaller and train driver so as to ensure/continue railway operation in specific situations.

Note 1: Operational instructions exist at both the national and European level.

[*2019/773/EU - TSI OPE - Appendix J - Glossary*]

20. Payload

The core content of a message, containing the valuable information intended for use by the recipient.

Note 1: In the context of the sending of Digital Instructions, the payload can contain one or more operational instructions.

21. Roster

A list or plan indicating the duty or leave schedule for individuals, groups, or physical resources within an organisation.

Note 1: In railway operations, rosters may apply to drivers, conductors, or rolling stock.

22. Segment Profile (SP)

A set of predominantly static infrastructure data required by the DAS on board (SFERA-OB/DAS) to compute the operational speed profile.

Note 1: Derived from *ERA Railway Terminology Collection*.

23. Timing point (TP)

A location identified in a train's schedule for which a specific time is assigned.

Note 1: The Timing point may correspond to an arrival time, a departure time, or a passing time of a train not scheduled to stop.

Note 2: Derived from the *ERA Railway Terminology Collection*.

24. Time window

A time interval within which a train is required to stop, start, or pass at a given location.

Note 1: Time windows for timing points are set in the JP.

25. Traffic Management System (TMS)

A non-safety-critical system that provides recommendations to the control-command and signalling system based on the current traffic situation and other internal or external information, in order to improve the level of service.

[*ERA Railway Terminology Collection*]

Note 1: In this IRS, a TMS encompasses all systems used by the infrastructure manager to manage traffic.

26. Train Characteristics (TC)

A set of predominantly static train data required by the DAS on board (SFERA-OB/DAS) to calculate the operational speed profile.

Note 1: TCs may be:

- Provided by the trakside via SFERA
- Provided directly by the TCMS
- Stored on the SFERA-OB/DAS
- Manually provided by driver input

27. Train path

The infrastructure capacity needed to run a train between two places over a given period.

[*2012/34/EU - Art. 3*]

28. Train path envelope (TPE)

A dataset specifying, for each location, a time window and, optionally, minimum and maximum speeds.

Note 1: The train path envelope is defined by the infrastructure manager's traffic management system or personnel, and is based on the scheduled timetable and line-specific traffic regulation. It is used as the basis for the JP.

29. Transmission System Operator (TSO)

The natural or legal person responsible for operating, ensuring the maintenance of and, if necessary, developing the transmission system in a given area and, where applicable, its interconnections with other systems, and for ensuring the long-term ability of the system to meet reasonable demands for the transmission of electricity.

[*2009/72/EU - Art. 2*]

Note 1: Transmission refers to the conduction of electricity via the extra high-voltage and high-voltage interconnected systems for delivery to end users or distributors, but does not include supply.

30. Verification

The confirmation, through examination or the provision of objective evidence, that specific requirements have been met. Objective verification demonstrates that, for specific inputs, the deliverables meet in all requirements of the relevant stage.

Note 1: Based on *EN 50126* - Definitions

Note 2: Verification is conducted at various lifecycle stages, assessing the system and its components to determine conformity with the requirements defined for each stage.

31. Virtual balise

A *point on the railway network for which both the exact location on an SP (distance from the start of a segment) and the geographical coordinates (latitude, longitude, and optionally altitude) are known*.

Digitalisation, Data, Emerging Innovations
Exchange of Data
Data exchange with Driver Advisory Systems (DAS) following the SFERA protocol

General Application

1 – Introduction

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

1.1 – General

To address the challenges posed by intermodal competition, the railway sector needs to focus on enhancing the overall performance of its rail system. Railway undertakings (RUs) and infrastructure managers (IMs) are therefore creating solutions to optimise the use of time slots, with a strong emphasis on punctuality, energy efficiency, and minimising time loss due to unforeseen procedures, while also ensuring the interoperability of cross-border traffic. Among other promising functions to facilitate train operation communication between ground and board, Driver Advisory Systems (DAS), Digital Instructions (DIs) and Free Text Communication (FTC) offer significant improvements in the short term without the need for large-scale investment.

- DAS are among the most effective methods for reducing traction energy use, with savings estimated between 5% and 15%[1] depending on different parameters (such as whether the DAS only uses preplanned path data or connects to the Traffic Management System (TMS) to obtain traffic management updates while running).

- DIs facilitate the communication of operational instructions by replacing verbal communication and/or physical handovers.

- FTC enables multilingual text-based information exchange between the train driver and his operational contact points on the ground, allowing the exchange of messages while reducing the level of language proficiency needed when driving a train on a network using a different language.

As described in this IRS, the SFERA protocol defines technical guidelines to handle train operation data exchange in terms of DAS, DI, and FTC functionalities. These are considered as functions within the SFERA protocol. It also facilitates the use of these functions during cross-border journeys and, where applicable, enables their connection to IM business tools such as TMSs.

As the implementation of train operation data exchange for DAS, DI, and FTC with the SFERA protocol may be realised as a single or separate solutions, the SFERA protocol defines general as well as specific requirements for all functions.

The exchange of train operation data is not an end in itself, but rather a tool for achieving specific business outcomes. Additionally, systems such as DAS and Automatic Train Operations (ATO) can contribute to saving energy, improved punctuality, additional network capacity, power management, enhanced disruption management, improved safety, and reduced carbon emissions.

In the context of delivering operational instructions, DIs play a key role in enhancing the quality of safety-related communication, minimising the likelihood of verbal communication errors, and reducing time losses in communication. On the other hand, FTC facilitates communication between signallers and drivers, particularly in multilingual environments, ensuring clarity, traceability, and operational efficiency.

1. Experiments at SNCF in 2018 on intercity trains showed a 13% reduction in energy use.

Fig. 1 - page 21 shows the relevant steps from the data being transmitted using SFERA to the business outcome.

This IRS also defines the requirements that help achieve cost-effective ways to implement the entire system (software, hardware, and connectivity).

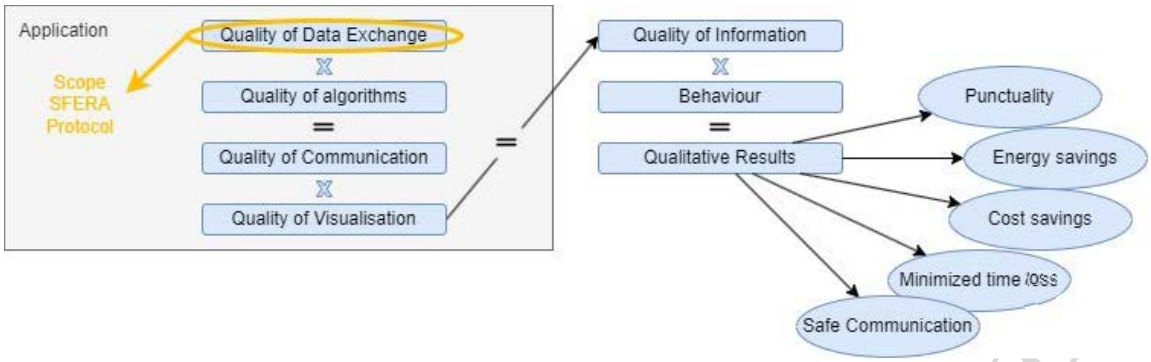


Fig. 1 - Steps from data transmission to business outcome

The reasons for which RUs and IMs may wish to implement a SFERA function vary.

The SFERA protocol establishes a common basis for the exchange of operational information in a multi-RU environment, enabling the development of interoperable and vendor-independent on-board and trackside systems. It supports seamless integration between different IM systems, which may rely on a particular business tool, such as a particular TMS.

The adoption of one or more harmonised protocol functions by all railway stakeholders offers the following benefits:

1. **Freedom of choice for RUs**: Railway undertakings can select their preferred products, which compete on functionality, efficiency, and effectiveness in a fair and open market, without being “locked in” on a single solution.

2. **Improved cross-border operations**: RUs can enhance their business processes by ensuring that their systems operate seamlessly cross border.

3. **Simplified interfaces for IMs**: Infrastructure managers can provide a single, standardised interface to all RUs, regardless of the specific systems they employ.

4. **Facilitated migration**: Transitioning between different products, such as different Train Protection Systems (TPSs) or ATO systems, becomes easier and more streamlined.

5. **Enhanced system consistency**: Greater alignment in the solutions adopted by RUs and IMs, minimising incompatibilities between systems to the greatest possible extent.

The IRS also reflects recent developments in the application of TSIs in Europe:

- CCS TSI: The implementation of Connected DAS serves as a preparatory step towards the broader deployment of ERTMS/ATO by defining data exchange flows and communication systems between on-board and ground systems. The ERTMS/ATO system under the CCS TSI has standardised data exchange through SUBSET-126.
  SFERA is defined as a superset of SUBSET-126, enabling ERTMS/ATO data to be exchanged using the SFERA interface. Every effort was made when developing this IRS to align with the ERTMS/ATO specifications to facilitate the transition from one configuration to another.

- TAF/TAP TSI: IMs and RUs are increasingly adopting the *TAF/TAP TSI* data exchange protocol, especially for planning and operating trains. The SFERA protocol incorporates elements from *TAF/TAP TSI* wherever possible, enabling stakeholders to share and reuse systems and functionalities.

- OPE TSI: IMs and RUs are required to implement Appendix A (*ERTMS operational principles and rules*) and Appendix C2 (*European Instructions*). This necessitates the use of European Instructions in line with the ERTMS operational principles and rules. The SFERA protocol enables the digital transmission of both European and National Instructions.

Close attention was also paid to clearly distinguishing between the safety functions of DAS and ATP. While DAS may indirectly enhance safety by reducing the number of restrictive signals encountered, the safety requirements and processes related to safety are handled by the ATP systems. Signal status and speed restriction information is intended solely for prediction purposes.

The same principle also applies to the DI and FTC functions within the train operation data exchange protocol. While the digital communication of operational instructions may have a positive influence on safety procedures, the scope and content of the European or National Instructions must remain governed by the TSI OPE and the applicable regulations for the relevant IM/RU.

NB: In this IRS, RU refers specifically to the “responsible” RU (elsewhere known as RRU), as defined in the TAF/TAP RU/IM Telematics Sector Handbook, i.e. the entity responsible for the operation of the train during the journey or a defined section thereof.

1.2 – System overview

A SFERA protocol-compliant environment for train operation data exchange is composed of various components, including:

- **Ground based components**: For example, a TMS

- **Communication infrastructure**: Responsible for facilitating data exchange between the ground and a train

- **On-board components**: For example, a business tool used by the train driver

The primary objective of the SFERA protocol is to specify the interfaces between the various systems involved in IM-Driver communication.

Fig. 2 - page 23 provides a comprehensive visual representation of all of the systems, including their interfaces and whether they concern the IM or RU. Interface specification boxes indicate whether the respective specifications are standardised within the SFERA protocol or left to the railway to

define, although SFERA could also be used for other interfaces. The SFERA protocol also supports different communication architectural setups, as described in point 1.3.1 - page 25.

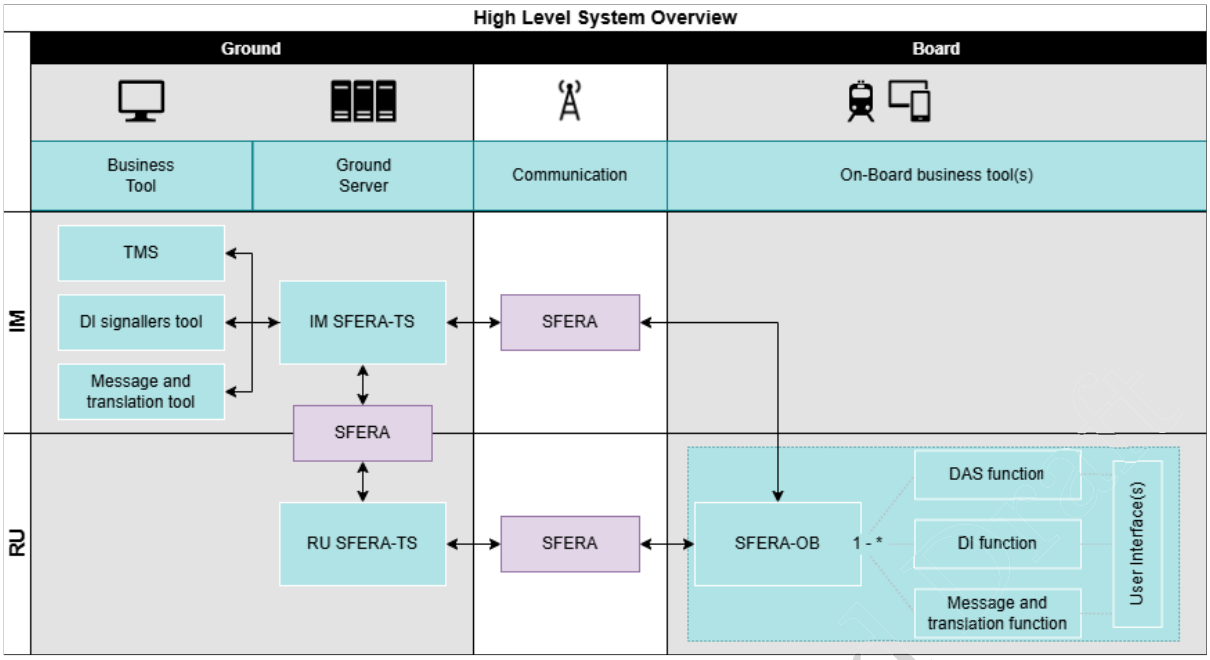


Fig. 2 - High-level system overview and SFERA-applicable interfaces

For DIs, it is essential to ensure that the data elements defined in the protocol are clearly presented to the driver. Therefore, SFERA harmonises instruction formatting on the driver's user interface.

Ground (or trackside) systems include:

- **The TMS**: Operated by the IM, it provides the data required to operate a DAS and calculates updated train timetables. The TMS then transmits this information to the IM SFERA-TS, typically in a specific format defined by the IM or TMS provider.
  In this IRS, the term "TMS" is used in its broadest sense: it represents all of the systems needed by the IM to manage traffic, including infrastructure data (e.g. gradients, virtual or physical balise positions), power management, rolling stock limitations (e.g. braking), and environmental factors (e.g. adhesion) that may not be present in some Traffic Management Systems.

- **DI signaller tool**: A software solution operated by the IM for creating and managing European and National Instructions in accordance with the applicable business rules and legal provisions.

- **Message and translation tool**: A software solution operated by the IM to send, receive, and display text messages to and from the driver. It is particularly useful in multilingual contexts where drivers may have limited knowledge of the local operating language(s), and can offer translation:

  - From the IM's operating language to other languages familiar to the driver
  - To the IM's operating language, allowing IM staff to understand driver messages written in other languages

- **IM SFERA-TS**: The IM SFERA Trackside Module establishes the communication with the RU SFERA-TS or directly with the SFERA On-Board Unit. It can also be integrated into the IM business tools (e.g. TMSs, DI signaller tools, etc.). It packages data from these tools into SFERA messages, and transmits to or receives data from the RU SFERA TS or OB.

- **RU SFERA-TS**: The RU SFERA Trackside Module is an optional intermediary server between the IM SFERA-TS and the trains' SFERA On-Board Unit. Where present, it establishes connections between the ground and on-board systems. For interoperability, it is recommended that the SFERA protocol be used for data exchange between the RU SFERA-TS and the SFERA On-Board Unit.

Communication infrastructure:

A unified data model is used across all communication links, though the communication layers differ depending on their characteristics:

- For server-to-server communication (between IM SFERA-TS and RU SFERA-TS), the communication layer to transmit the SFERA messages is established through the IM and RU's Common Interfaces.

- For the other links concerning communication with the SFERA-OB (between SFERA-OB and IM SFERA-TS/RU SFERA-TS), a specific SFERA communication layer has been designed, using standard protocols established for Internet of Things use cases (i.e. many end devices communicate with a central server which aggregates data).

SFERA is compatible with any network connection that supports TCP/IP (e.g. the Internet).

On-board systems:

The SFERA-OB (On-Board) application exchanges data with ground systems, and is used by the driver via a mobile device, or an integrated cab interface. It may host one or more of the following functions:

- **DAS On-Board**: Its primary role is to calculate, send, and then display the driving advice to the user interface (if not carried out by the RU or IM ground systems). It also collects train status data (e.g. position, speed) to send this back.

- **DI On-Board**: Its primary role is to receive the European Instructions and National Instructions sent by the IM, as well as handle the required transactions and status updates.

- **FTC On-Board**: Its primary role is to transmit and display messages written by the driver for the ground staff and systems, and vice versa.

A user interface is required for DAS, DI and FTC functions to display information to the driver and permit data entry.

Depending on the RU's implementation strategy and deployment constraints, the SFERA functions and their user interfaces may be integrated into a single application or distributed across multiple devices (e.g. DAS on a tablet, DI and FTC on a smartphone).

1.3 – SFERA-compliant interfaces and architectural setups

The SFERA protocol can be implemented for all three interfaces specified in Fig. 2 - page 23. For each type of link, SFERA compliance may only be advertised if all layers of the protocol are implemented according to the specifications outlined in this IRS. This ensures that stakeholders are correctly informed about the interoperability level provided by a specific implementation.

If an IM, RU, or supplier chooses to implement only parts of the protocol, they must clearly disclose these decisions. They shall be transparent about not using SFERA for all three interfaces or only partially implementing it (e.g. only the XSD between RU SFERA-TS and SFERA-OB).

1.3.1 – SFERA architectural setups

The SFERA protocol supports three possible architectural setups, each offering distinct mechanisms for exchanging train operation data between ground systems and on-board devices. These setups are designed to be compatible with various operational needs; however, the choice of setup and which functionalities are implemented, shall be defined and agreed upon by the involved parties on a case-by-case basis.

These setups are:

- **IM-RU**: Here, the SFERA protocol defines an interface specification between the IM SFERA-TS and the RU SFERA-TS. The RU SFERA-TS then communicates with the on-board devices, also using the SFERA protocol. For those links to be considered SFERA compliant, all protocol layers must be used. Note that the communication layers for the two links differ: the IM SFERA-TS ↔ RU SFERA-TS link uses the Common Interface, while the RU SFERA-TS ↔ SFERA-OB link uses a SFERA-specific communication layer. Both links share a common XML message definition.

- **IM-Train**: Here, the on-board SFERA device connects directly to the IM SFERA-TS. The SFERA protocol shall be used for this communication channel.

- **ERTMS/ATO**: Here, SFERA is used to connect the TMS to an ATO trackside system. This is possible because SFERA is a superset of ERTMS/ATO SUBSET-126. This setup may also simplify implementation on the IM side.

The actual setup used is situation specific. The aim is for SFERA to be compatible with all setups, but its implementation is left to the IM and RU. Therefore, although SFERA can be used in all three setups, this does not imply that all functionalities must be mandatorily supported in every case. The functionalities to be used and/or supported need to be defined and agreed upon on a case-by-case basis by the parties involved.

The setups are described in detail in the following sections.

1.3.1.1 – IM-RU setup: “Back-office” communication between the IM SFERA-TS and RU SFERA-TS

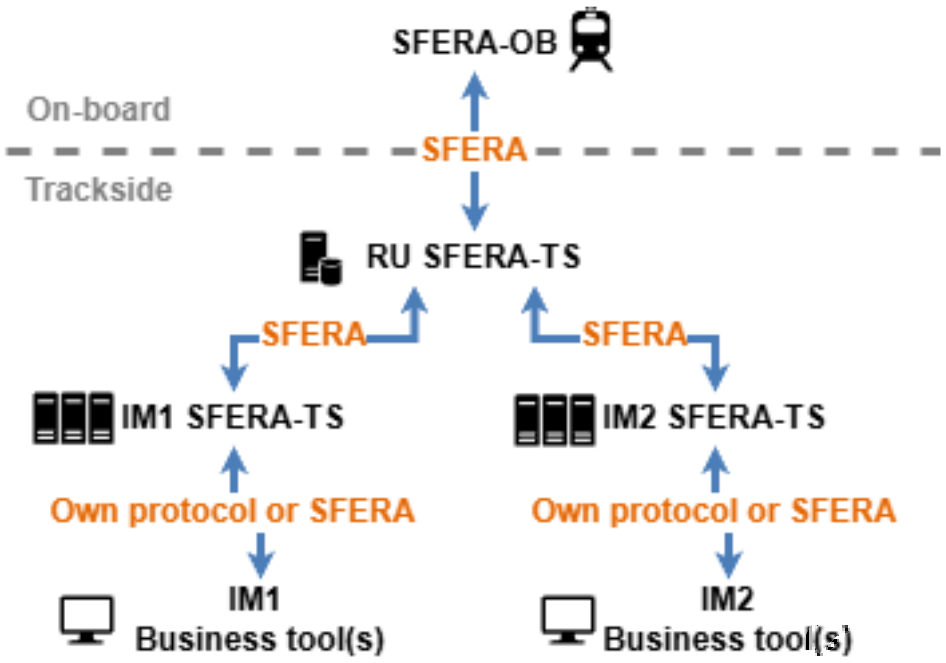


Fig. 3 - Connection through RU SFERA-TS

In this setup, the IM SFERA-TS only interacts with the RU SFERA-TS. It is up to the RU to act as a gateway and relay the information between the IM systems and each SFERA-OB. This configuration may be chosen if:

- The RU requires access to the data sent to the on-board devices for business analysis.
- The IM prefers not to establish direct connections with RU devices (e.g. for cybersecurity reasons).

The advantages of this setup are that it gives the RU influence over the devices and allows it to add additional functionality (e.g. if the RU itself is also interested in the information sent from the OB device for its own business processes).

As part of this setup, it is important that the parties involved reach an agreement, particularly regarding information quality and guaranteed delivery.

In this case, the SFERA protocol is applied:

- Between the IM and RU

- Between the RU and SFERA-OB. This is recommended for interoperability purposes, especially for RUs operating on networks that do not implement the IM-RU setup.

The communication layer will be different in the two links:

- IM SFERA-TS <-> RU SFERA-TS: SFERA messages shall be exchanged through the IM and RU Common Interfaces.

- RU SFERA-TS <-> SFERA-OB: SFERA messages shall be exchanged through a communication layer designed specifically for SFERA, based on standard technologies for server-to-mobile-device communication.

1.3.1.2 – IM-Train setup: Direct communication between the IM trackside and on-board systems

This setup is a simplified version of the first setup given in point 1.3.1.1 - page 26. In this case, the on-board device communicates directly with the IM systems responsible for the area in which the train is operating. No ground gateway server is required. However, without it, the available functionality is limited to that supported by SFERA. For example, the RU cannot get a copy of the data returned from the on-board device.

Therefore, this setup might be preferred, for instance, by RUs that have a smaller fleet, or by IMs that favour minimum latency.

This setup may also be used for ATO over Class B system applications.

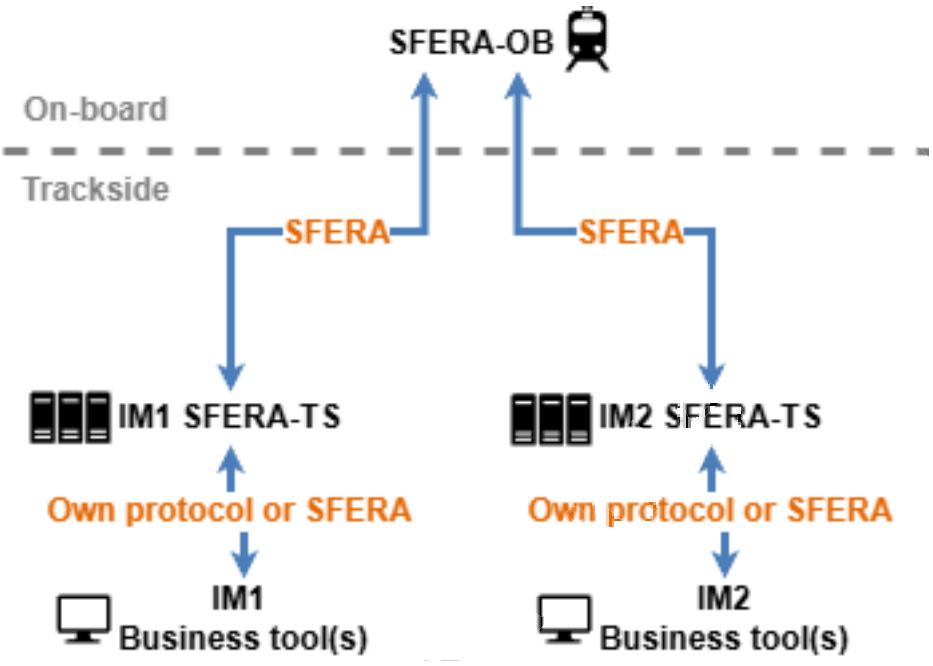


Fig. 4 - Direct connection between IM and on-board

The communication layer that is used in this setup is identical to the RU SFERA-TS <-> SFERA-OB link of the IM-RU setup.

1.3.1.3 – ERTMS/ATO setup

In this setup, SFERA is used in the context of ERTMS/ATO. It is designed to be compatible with the ERTMS/ATO standard (specifically SUBSET-126). The information required for SUBSET-126 is a subset of the data transferred via the SFERA protocol. As a result, it is possible to use the SFERA protocol to connect the TMS to ATO trackside equipment.

This may be of interest to IMs aiming to support both DAS and ERTMS/ATO configurations. While the use of SFERA is not mandatory for connecting an ATO trackside system, it offers a practical option.

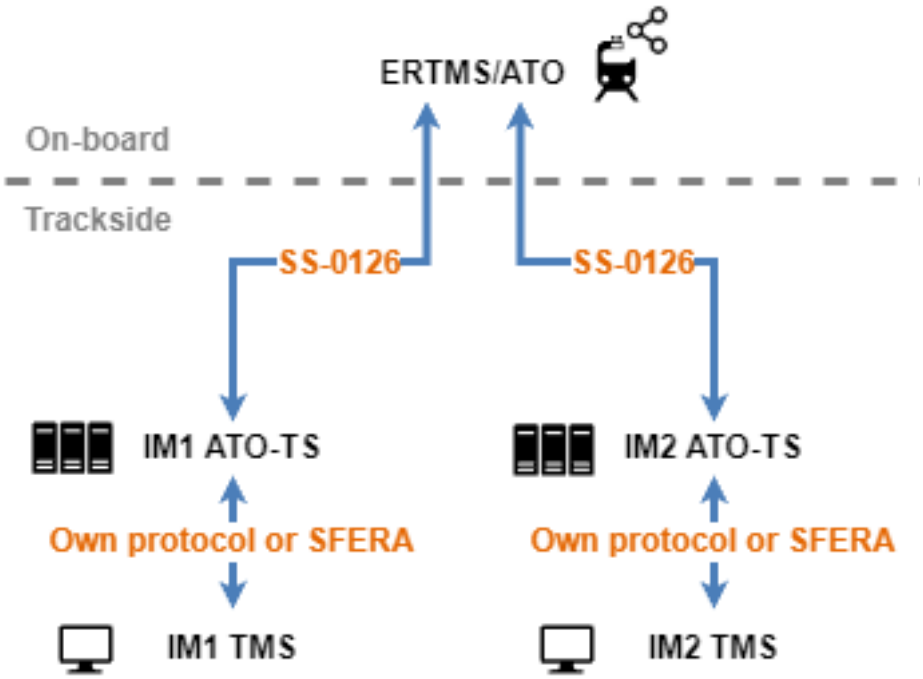


Fig. 5 - ERTMS/ATO connection

2 – Requirements of the exchange protocol

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

2.1 – Introduction to SFERA requirements

The requirements for SFERA are derived from the key business objectives of punctuality, energy and cost savings, and enhanced safety, but they only represent a part of how to achieve them.

SFERA requirements fall into several categories, each of which is elaborated in the following sections:

- Interoperability (see point 2.2 - page 30)
  To achieve reduced energy consumption, safety improvements and punctuality in a cost-effective manner, the SFERA protocol shall ensure the interoperability of DAS and ATO solutions.

- Evolution (see point 2.3 - page 33)
  To obtain results quickly and with limited investment, the protocol shall support solutions that are able to evolve over time.

- Setups (see point 2.4 - page 36)
  The protocol shall support implementation in various architectural setups, depending on local conditions and existing systems.

- Communication (see point 2.5 - page 37)
  The communication protocol shall ensure the reliable and uncorrupted transfer of data to the appropriate parties or systems.

- Integrity (see point 2.6 - page 38)
  The SFERA protocol shall include mechanisms to guarantee data authenticity and integrity.

- Data (see point 2.7 - page 39)
  The functional requirements regarding the types of data to be exchanged are defined by current and anticipated use cases for functions used by drivers (e.g. DAS, ATO, DI, multilingual message exchange).

- Maintainability (see point 2.8 - page 41)
  As not all future developments can be predicted, the protocol shall be designed with flexibility in mind, ensuring long-term maintainability.

2.2 – Interoperability requirements

To reduce costs and encourage the widespread adoption of connected digital functions for drivers, the SFERA protocol shall facilitate interoperability. Interoperability enables solutions to function:

- Across (IM) borders
- Across different ATP systems
- From ATO to non-ATO areas
- On both TMS-supervised and non-TMS-supervised tracks

Regardless of where trains operate, they can all benefit from functions that communicate with IMs using the same data exchange protocol. Without such interoperability, each RU-IM combination would need to develop their own proprietary solution, increasing costs.

As illustrated in Fig. 6 - page 31, trains shall be able to operate along all such routes using the same data exchange protocol.

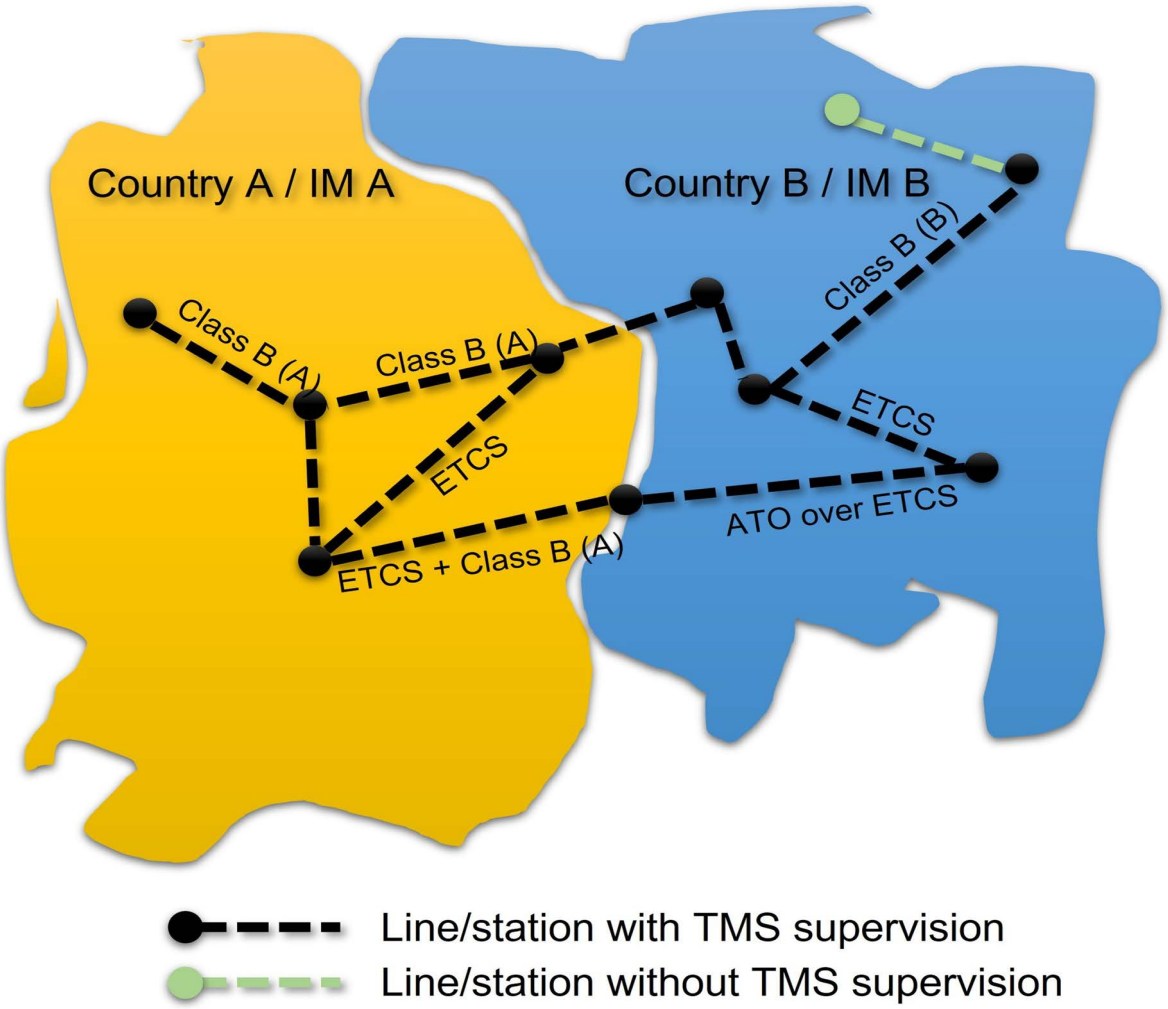


Fig. 6 - Lines with different ATPs crossing IM borders

Table 1: Interoperability requirements

ID	Requirement	DAS	DI	FTC
IO_REQ_001	The SFERA protocol shall support the exchange of on-board (RU) and ground (IM) data so that the DAS is able to establish driving advice for the driver.	Yes		
IO_REQ_002	The SFERA protocol shall ensure the seamless transition of on-board data to the ground while the train moves from an area managed by one IM to an area that is managed by another IM.	Yes	Yes	Yes
IO_REQ_003	The SFERA protocol shall include all of the data from ERTMS/ATO SUBSET-126.	Yes		
IO_REQ_004	The SFERA protocol shall facilitate the exchange of on-board to ground data while the train moves from one ATP system to another.	Yes	Yes	Yes

Table 1: Interoperability requirements

ID	Requirement	DAS	DI	FTC
IO_REQ_005	The SFERA protocol shall facilitate the exchange of on-board to ground data while the train moves from an area where it is manually operated to an area where it is automatically operated and vice versa.	Yes		
IO_REQ_006	The SFERA protocol shall not implement new data interfaces for data exchange use cases/processes already covered by TSI data interfaces.	Yes	Yes	Yes
IO_REQ_007	The SFERA protocol shall be able to provide advisory information for traffic management areas and non-traffic management areas. In unmanaged areas, the DAS will operate as an S-DAS.	Yes		
IO_REQ_008	The SFERA protocol shall preferably use international or widely used standards (e.g. Telematics TSI).	Yes	Yes	Yes
IO_REQ_009	The SFERA protocol shall be supplier-agnostic in order to achieve interoperability.	Yes	Yes	Yes
IO_REQ_010	The SFERA protocol shall not implement new technical interfaces between RU DAS-TS and IM DAS-TS that are already covered by Telematics TSI specifications.	Yes	Yes	Yes
IO_REQ_011	The SFERA protocol shall be able to digitally exchange DIs on networks that have a ground DI function.		Yes	
IO_REQ_012	The SFERA protocol shall permit the driver to send back findings reports requested by the signaller.		Yes	
IO_REQ_013	The SFERA protocol shall allow the delivery of Digital Instructions while the train is on the preceding IM's network.		Yes	
IO_REQ_014	The SFERA protocol shall allow the delivery of a findings report once the driver has left the IM's network.		Yes	
IO_REQ_015	The SFERA protocol shall include all of the instructions defined in the OPE TSI.		Yes	
IO_REQ_016	The SFERA protocol shall permit the ground to know if a DI function has been activated by the driver.		Yes	
IO_REQ_017	The SFERA protocol shall permit the transmission of instructions specific to IM regulations, or extensions of OPE TSI instructions (e.g. National Instructions).		Yes	
IO_REQ_018	The SFERA protocol shall permit the transmission and presentation of the instructions in multiple languages.		Yes	
IO_REQ_019	The SFERA protocol shall propose a solution for sharing the structure and formatting of instructions, which can be preloaded on the on-board DI device.		Yes	
IO_REQ_020	The SFERA protocol shall permit the cancellation of an instruction that has not yet been accessed by an RU.		Yes	
IO_REQ_021	The SFERA protocol shall permit the revocation of an instruction that has already been accessed by an RU.		Yes	

Table 1: Interoperability requirements

ID	Requirement	DAS	DI	FTC
IO_REQ_022	The SFERA protocol shall permit the ground to inform the board of the parts of the network on which the DI function is supported.		Yes	
IO_REQ_023	The SFERA protocol shall permit the ground to inform the board of the parts of the network on which the DI function is temporarily unavailable.		Yes	
IO_REQ_024	The SFERA protocol shall cover the eventuality of an instruction received by the device, but read by the driver in a zone without data connection.		Yes	
IO_REQ_025	There shall be a process defining when a fallback procedure is required through the SFERA protocol, and proposing a fallback procedure in such cases (e.g. loss of connection, hardware failure).		Yes	
IO_REQ_026	The SFERA protocol shall permit steps of the procedure to be skipped, if some of them are completed offline.		Yes	
IO_REQ_027	The SFERA protocol shall define what happens when a message has not been acknowledged.		Yes	
IO_REQ_028	In the case of connectivity loss, the SFERA protocol shall permit a procedure to be resumed, without starting over.		Yes	

2.3 – Evolution requirements

In order to achieve cost-effective benefits in terms of punctuality and energy savings, the quality of advice to the driver depends not only on the quality of the algorithms, but also on the quality of the data. To realise the benefits as soon as possible, only a limited number of elements are mandatory to begin a DAS implementation.

It is also possible to start with a less accurate dataset which improves over time. This approach lowers the threshold for initial deployment, and by using the SFERA protocol, future evolution is possible. Moreover, the lower bandwidth requirements associated with less accurate data may justify lowering the quality on certain parts of the network.

The greater the availability and quality of data, the more accurately the DAS or ATO will be able to support trains in running both on time and with lower energy consumption (e.g. it shall be possible to start with a basic DAS and progressively add new data elements, such as gradients, to the calculations) without needing to redesign the entire data exchange.

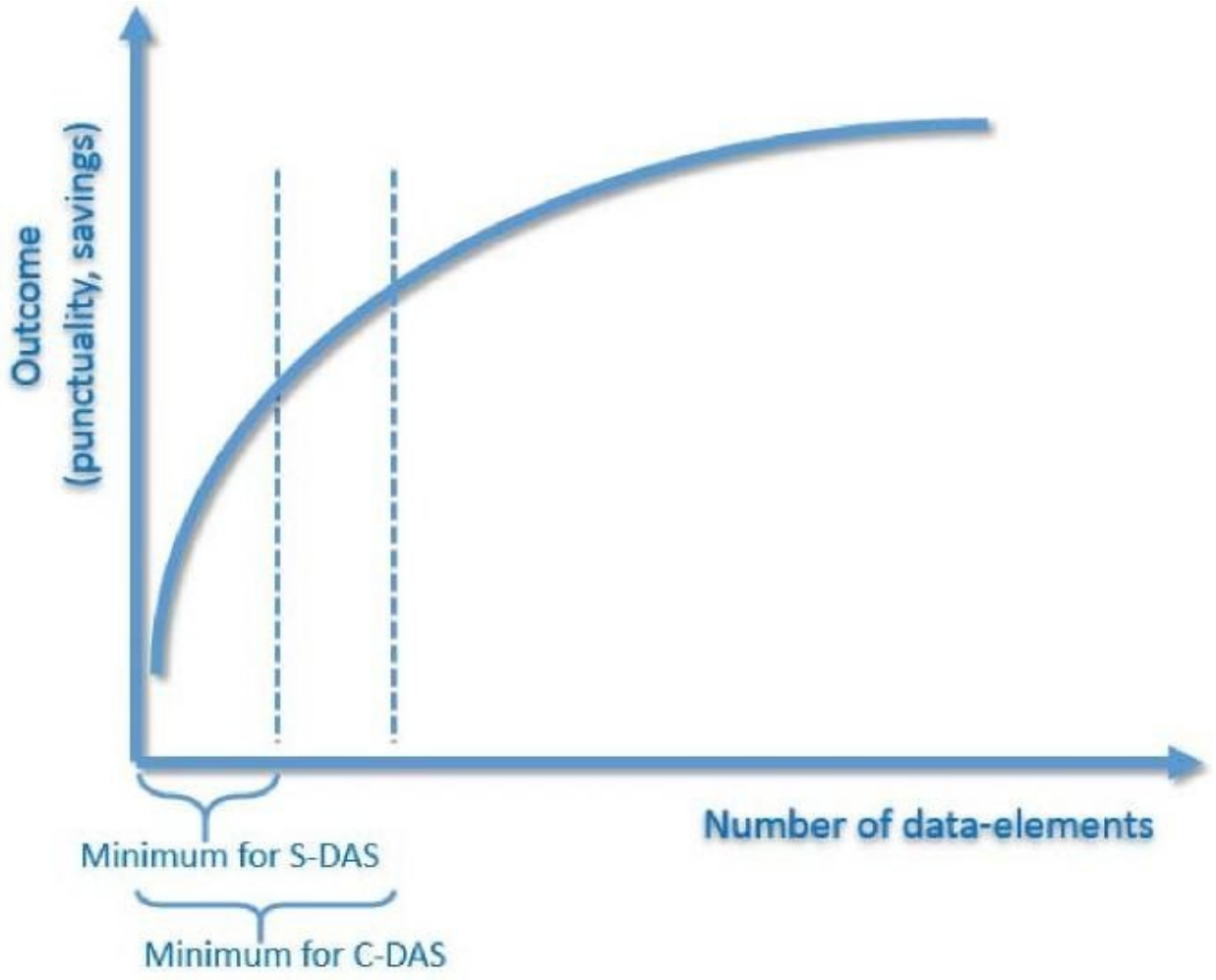


Fig. 7 - Outcome as a function of the number of data elements

This requirement, alongside interoperability, necessitates the inclusion of indicators on data quality (e.g. precision, timeliness) within the SFERA data protocol.

The same principle applies for the DAS and ATO systems: cost-effective implementations may use fewer data elements, even if more comprehensive data is available in some of the regions where DAS is deployed.

NB: As the SFERA protocol does not enhance data, it is important that procedures for preparing and acquiring data are established to reduce the risk of incorrect data being used by the system to an acceptable level.

Table 2: Evolution requirements

ID	Requirement	DAS	DI	FTC
EV_REQ_001	It shall be possible to use SFERA even when not all possible data elements are used.	Yes	Yes	Yes
EV_REQ_002	Data elements for the correct functioning of the protocol and communication shall be mandatory.	Yes	Yes	
EV_REQ_003	The SFERA protocol shall only allow the required data to be transferred.	Yes	Yes	Yes
EV_REQ_004	The SFERA protocol shall support both offline (S-DAS) and connected (C-DAS) options.	Yes		
EV_REQ_005	The SFERA protocol shall define the minimum dataset required for proper operation.	Yes	Yes	Yes
EV_REQ_006	The SFERA protocol shall permit the DAS, DI, and FTC functions to run as distinct applications on the same device (distinct application layers).	Yes	Yes	Yes
EV_REQ_007	The SFERA protocol shall permit the DAS, DI, and FTC functions to run on the same or different devices.	Yes	Yes	Yes
EV_REQ_008	The SFERA protocol shall be implementable on both mobile devices (e.g. smartphone, tablet) and static platforms (e.g. a device integrated in the train).	Yes	Yes	Yes
EV_REQ_009	The SFERA protocol shall permit the implementation of digital findings outside the SFERA protocol.		Yes	
EV_REQ_010	The SFERA protocol shall permit the implementation of instruction confirmation from the driver outside the SFERA protocol.		Yes	
EV_REQ_011	The SFERA protocol shall permit the implementation of location verification outside the SFERA protocol.		Yes	
EV_REQ_012	The SFERA protocol shall implement common solutions for shared functionalities across the different functions: communication layers, handshake, driving modes, journey and infrastructure descriptions.	Yes	Yes	Yes

2.4 – Requirements supporting different setups

Since different implementations exist and not all IMs and RUs are the same, SFERA shall facilitate different setups: C-DAS-O (default) and C-DAS-C (alternative for transition purposes), as well as the different architectural setups to exchange SFERA data between an on-board system and the ground.

Table 3: Requirements supporting different setups

ID	Requirement	DAS	DI	FTC
SU_REQ_001	The SFERA protocol shall support the exchange of data for ERTMS/ATO SUBSET-126.	Yes		
SU_REQ_002	The SFERA protocol shall support the different communication setups: IM-RU, IM-Train and ERTMS/ATO (as described in point 1.3.1 - page 25).	Yes		
SU_REQ_003	The SFERA protocol shall support bi-directional communication between IM SFERA-TS and RU SFERA-TS, IM SFERA-TS and SFERA-OB, and RU SFERA-TS and SFERA-OB.	Yes	Yes	
SU_REQ_004	The SFERA protocol shall not interfere or conflict with other technologies such as ATO or ERTMS.	Yes		
SU_REQ_005	The SFERA protocol shall enable Timetable, S-DAS, C-DAS-O and C-DAS-C solutions (as described in point 3.1 - page 43).	Yes		
SU_REQ_006	The SFERA protocol shall support manual input of the train's and signaller's location.		Yes	
SU_REQ_007	The SFERA protocol shall support security measures to ensure instructions are issued to the correct train driver.		Yes	
SU_REQ_008	The SFERA protocol shall permit DI exchange in the IM-RU and IM-Train setups.		Yes	
SU_REQ_009	The SFERA protocol shall permit plain text message exchange in the IM-RU and IM-Train setups.			Yes

NB: Although not every country/IM is obliged to support all of the setups, it is important to keep interoperability in mind when determining which will be supported. IMs and RUs are to make bilateral agreements to this effect.

2.5 – Communications requirements

The communication protocol shall enable a cost-effective and swift data transmission, ensuring that it is delivered uncorrupted and only to the designated parties while trains are in operation and crossing (IM) borders.

Table 4: Communications requirements

ID	Requirement	DAS	DI	FTC
CP_REQ_001	SFERA communication shall be fair and inclusive for both small and large RUs.	Yes	Yes	Yes
CP_REQ_002	SFERA communication shall be interoperable between different IMs.	Yes	Yes	Yes
CP_REQ_003	SFERA communication between ground and on-board systems shall be delivered as quickly as possible (within the limits of the network latency).	Yes	Yes	Yes
CP_REQ_004	SFERA communication between ground and on-board systems shall be encrypted.	Yes	Yes	Yes
CP_REQ_005	SFERA message delivery shall be guaranteed in both directions, including during all physical connectivity losses.	Yes	Yes	Yes
CP_REQ_006	SFERA communication shall transfer complete data objects rather than only differentials.	Yes	Yes	Yes
CP_REQ_007	SFERA communication shall enable DAS to work with the last JP received.	Yes		
CP_REQ_008	SFERA communication shall work on any TCP/IP-based network and may be based on the public Internet.	Yes	Yes	Yes
CP_REQ_009	SFERA communication shall not be accessible to the general public, but only to specific RUs (or RU-contracted entities) responsible for a given train.	Yes	Yes	Yes
CP_REQ_010	For the validity of B2G messages, the SFERA protocol shall provide a means of driver authorisation on hand-held devices or front cab instrumentation to initiate communication.	Yes	Yes	Yes
CP_REQ_011	SFERA communication shall allow multiple SFERA-OB instances of a function to be connected for the same train, with only one being active and able to send messages that have a functional impact on the IMs business functions.	Yes	Yes	Yes
CP_REQ_012	SFERA communication shall be platform-independent and vendor-independent.	Yes	Yes	Yes
CP_REQ_013	SFERA communication shall align with the current separation of responsibilities between RUs and IMs regarding train operation processes.	Yes	Yes	Yes
CP_REQ_014	Authentication and access management shall be independent of the rest of the SFERA protocol.	Yes	Yes	Yes
CP_REQ_015	SFERA communication shall be based on RUs being responsible for user and device management for their own personnel and equipment.	Yes	Yes	Yes

Table 4: Communications requirements

ID	Requirement	DAS	DI	FTC
CP_REQ_016	SFERA shall not impose a specific method for user and device management on RUs.	Yes	Yes	Yes
CP_REQ_017	SFERA communication shall function correctly at all times, regardless of time zone and system time differences between communicating systems.	Yes	Yes	Yes
CP_REQ_018	SFERA shall integrate the Common Interface as a communication layer for the IM DAS-TS to RU DAS-TS link, benefiting from the pre-established interoperability layer between IM and RU servers.	Yes	Yes	Yes
CP_REQ_019	A space-saving format shall be used.	Yes	Yes	Yes
CP_REQ_020	The SFERA protocol shall provide a means to confirm message delivery (technical acknowledgement).	Yes	Yes	Yes
CP_REQ_021	The SFERA protocol shall not prevent simultaneous communication of different functions on the same device, whether within the same application or not.	Yes	Yes	Yes
CP_REQ_022	SFERA shall define an interoperable method for authorising users/devices to exchange data with SFERA-TS servers.	Yes	Yes	Yes

2.6 – Data integrity requirements

To guarantee that end-user solutions operate well, there shall be mechanisms in the SFERA protocol to ensure data authenticity and integrity.

Table 5: Data integrity requirements

ID	Requirement	DAS	DI	FTC
DI_REQ_001	The SFERA protocol shall provide mechanisms to ensure that information is exchanged between the designated systems (see communication requirements).	Yes	Yes	Yes
DI_REQ_002	The SFERA protocol shall provide a means of acknowledging receipt of messages.	Yes	Yes	Yes
DI_REQ_003	The SFERA protocol shall provide a means of verifying the validity of messages (e.g. versioning).	Yes	Yes	Yes
DI_REQ_004	The SFERA protocol shall provide solutions for resending information if the data does not arrive correctly.	Yes	Yes	Yes
DI_REQ_005	The SFERA protocol shall ensure that information is not corrupted during transmission between IM SFERA-TS, RU SFERA-TS and SFERA-OB.	Yes	Yes	Yes
DI_REQ_006	The SFERA protocol shall provide the possibility to ensure that information is not corrupted during end-to-end transmission from the IM's business tool to the SFERA-OB, as a mandatory function for DI and optional for DAS and FTC.	Yes	Yes	Yes
DI_REQ_007	The SFERA protocol shall permit the sending of instructions even if C-DAS data cannot be generated.		Yes	

Table 5: Data integrity requirements

ID	Requirement	DAS	DI	FTC
DI_REQ_008	The SFERA protocol shall ensure that the correct message is delivered to the correct train.		Yes	
DI_REQ_009	The SFERA protocol shall be able to include the ID of the driver communicating as an optional field in B2G messages.	Yes	Yes	Yes
DI_REQ_010	The SFERA protocol shall assign a unique ID to each message and a unique ID to each instruction within a message.		Yes	
DI_REQ_011	It shall be possible to use consumer-grade devices to run apps supporting the protocol while ensuring safety, security and integrity requirements are met.	Yes	Yes	Yes

2.7 – Functional data requirements

To support the SFERA functions, SFERA systems require accurate and relevant functional data. This may include information about the planned train journey (such as Timing Point (TP) targets and location-specific requirements), TCs that influence its performance, and infrastructure features that impose operational constraints. Additionally, to optimise energy usage, manage power supply limitations, and maintain operational continuity in the event of power-related issues, the exchange of data concerning power consumption and restrictions is essential.

The exact data required for a given SFERA function is determined by the relevant use cases, as well as the capabilities of the implemented or planned applications, ensuring interoperability and effective communication across different business tools and stakeholders.

The data requirements from the use cases are given in Table 6.

Table 6: Functional data requirements

ID	Requirement	DAS	DI	FTC
DA_REQ_001	The SFERA protocol shall have a logical data structure that accommodates both mandatory and optional data.	Yes	Yes	Yes
DA_REQ_002	The SFERA protocol shall accommodate elements that influence the running of the train (e.g. the planned train path and TP, temporary (speed) constraints, etc.).	Yes		
DA_REQ_003	The SFERA protocol shall accommodate TCs (e.g. length, weight, engine power, etc.).	Yes		
DA_REQ_004	The SFERA protocol shall provide static data on track segments (e.g. gradients, curves, static speed profile, permanent signals, tunnels, etc.) that influence acceleration and deceleration.	Yes		
DA_REQ_005	The SFERA protocol shall support the transfer of power-related data (e.g. requests to limit consumption or feedback with expected offtake/regeneration).	Yes		
DA_REQ_006	SFERA shall support the transfer of data on external circumstances that influence acceleration and deceleration, such as weather conditions.	Yes		

Table 6: Functional data requirements

ID	Requirement	DAS	DI	FTC
DA_REQ_007	The SFERA protocol shall support the transfer of data to manage low-adhesion conditions.	Yes		
DA_REQ_008	The SFERA protocol shall support the transfer of pre-calculated speed advice (for C-DAS-C setup).	Yes		
DA_REQ_009	The standard data structure shall allow transmission of network-specific data in a generic format, to accommodate specificities that would burden the standard SFERA model.	Yes	Yes	
DA_REQ_010	The SFERA protocol shall support the trackside in supplying the measured position of a train and nearby trains within the rail network.	Yes		
DA_REQ_011	The SFERA protocol shall support the transfer of additional context or procedural information to a driver that does not directly influence acceleration or deceleration.	Yes		
DA_REQ_012	The SFERA protocol shall support a train in reporting various aspects of its status to the trackside.	Yes		
DA_REQ_013	The SFERA protocol shall support the sharing of status updates for instruction delivery in both normal and degraded situations, following a harmonised workflow (status list and sequence).		Yes	
DA_REQ_014	The SFERA protocol shall permit the sharing of status updates for instructions from the signaller to the driver, with confirmation and possible extension of the instruction.		Yes	
DA_REQ_015	The SFERA protocol shall allow the SFERA-TS to identify the currently active SFERA function and the current driving mode for the driver.	Yes	Yes	Yes
DA_REQ_016	The SFERA protocol shall allow more than one instruction to be sent within one instruction payload, including their order of execution. All instructions shall be confirmed together.		Yes	
DA_REQ_017	The SFERA protocol shall handle changes in the regulations that define the instructions.		Yes	
DA_REQ_018	The SFERA protocol shall allow the Trackside DI function to know the status of its counterpart (readiness to exchange).		Yes	
DA_REQ_019	The SFERA protocol shall allow the DAS function to receive an instruction to disable advice for a specific track section (avoiding conflicting data on the DMI between DI and DAS functions).	Yes	Yes	
DA_REQ_020	The SFERA protocol shall allow the driver who handled the Digital Instruction to be identified.		Yes	
DA_REQ_021	The SFERA protocol shall ensure that the instruction's text and structure is maintained during transmission, while allowing customisation of the format.		Yes	
DA_REQ_022	The SFERA protocol shall permit a list to be sent of ground contact points with which Digital Instructions and plain text messages can be exchanged.		Yes	Yes
DA_REQ_023	The SFERA protocol shall allow the sending of plain text messages to the train driver, or a ground contact.		Yes	

Table 6: Functional data requirements

ID	Requirement	DAS	DI	FTC
DA_REQ_024	The SFERA protocol shall allow the receiver of a plain text message to specifically respond to that message.		Yes	
DA_REQ_025	The SFERA protocol shall allow the current content and status of Digital Instructions to be shared with Read-Only and inactive SFERA-OB devices.	Yes		
DA_REQ_026	The SFERA protocol shall allow the plain text messages exchanged between ground and active driver to be shared with Read-Only and inactive SFERA-OB devices.		Yes	
DA_REQ_027	The SFERA protocol shall allow any text information in different languages to be shared.	Yes	Yes	Yes
DA_REQ_028	The SFERA protocol shall allow the content of an instruction written by the driver in his DI function to be sent to the RU SFERA-TS.		Yes	

2.8 – Maintainability requirements

The SFERA protocol is based on best practices for current implementations of DAS and DI functions, and use cases planned for the near future.

Over time, new functions will be needed, new versions of ERTMS/ATO SUBSETS will be published, and therefore the SFERA protocol will need to be updated. Therefore, RUs and IMs should avoid customising the SFERA data structure. To meet new challenges that IMs and RUs are facing, and at the same time ensure interoperability and reduce costs through standardisation, SFERA shall include the appropriate procedures.

To stay flexible and interoperable over time, SFERA shall also be designed in such a way that flexibility takes precedence over the minimisation of data, although messages should still be kept as small as possible to enable quick transmission.

Table 7: Maintainability requirements

ID	Requirement	DAS	DI	FTC
MA_REQ_001	To maintain the SFERA protocol, procedures shall be established to ensure periodic, controlled updates.	Yes	Yes	Yes
MA_REQ_002	To maintain the SFERA protocol, procedures shall be established to ensure that SFERA stays compatible with ERTMS/ATO SUBSET-126.	Yes		
MA_REQ_003	Implementations of SFERA shall be able to handle minor and major SFERA updates.	Yes	Yes	Yes
MA_REQ_004	The SFERA protocol shall, for flexibility, accommodate “network specific parameters” to be sent from G2B and B2G within its structure to account for unforeseen specific use cases not covered by the “structured” part of the protocol.	Yes		

Table 7: Maintainability requirements

MA_REQ_005	SFERA shall be easy to expand with new data elements in future versions during the maintenance phase of this IRS.	Yes	Yes	Yes
MA_REQ_006	SFERA messages shall be easy to validate against the standard.	Yes	Yes	Yes
MA_REQ_007	To maintain the SFERA protocol, procedures shall be established to ensure that SFERA remains compatible with the Telematics TSI.	Yes	Yes	Yes
MA_REQ_008	Different implementations of network-specific parameters shall be periodically reviewed to assess whether the protocol needs to be modified.	Yes		
MA_REQ_009	Administrative overhead shall be kept low, e.g. minimal: - User administration - Metadata	Yes	Yes	Yes

3 – Exchange protocol functions

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

The following sections describe the available functions for which data can be exchanged with the SFERA protocol.

3.1 – Driver Advisory Systems (DAS)

A Driver Advisory System (DAS) assists drivers with the performance of a train service. The information is presented to the driver via an in-cab display (integrated into the rolling stock or in the form of a portable device).

The DAS receives data from different sources and calculates the ideal driving profile to fulfil the transport plan (and TPEs), while optimising the journey according to different criteria such as reduced energy consumption. Additionally, the DAS automatically recalculates the driving profile based on available updates to account for running late or early compared to the timetable and then advises a revised driving profile.

For example, if a train is running early, its DAS may display a speed lower than the speed limit in order to prevent unnecessary energy consumption and brake applications. In contrast, if a train is running late, its DAS may display the maximum possible speed, while respecting the constraints (train speed limits, environment, conflicts, etc.), to reduce the delay.

Speed advice is determined using a combination of two complementary running time calculations known as the “inner and outer control loops” (see also Fig. 8 - page 44):

- “Outer loop”: The **TMS establishes the optimal TPEs** for all trains, based on timetables, positions, TCs and estimated speeds, in order to avoid conflicts, ensure punctuality and optimise capacity usage. The TPE then defines windows for TPs to allow for live speed optimisation. These are sent to the DAS (or ATO).

- “Inner loop”: **The DAS (or ATO) of a specific train calculates the optimal speed within the TPEs** to optimise the train’s performance. The DAS (or ATO) sends updated TCs, position, and Status Reports (SR) (e.g. if the TPE cannot be followed) back to the TMS, which are used for the outer loop.

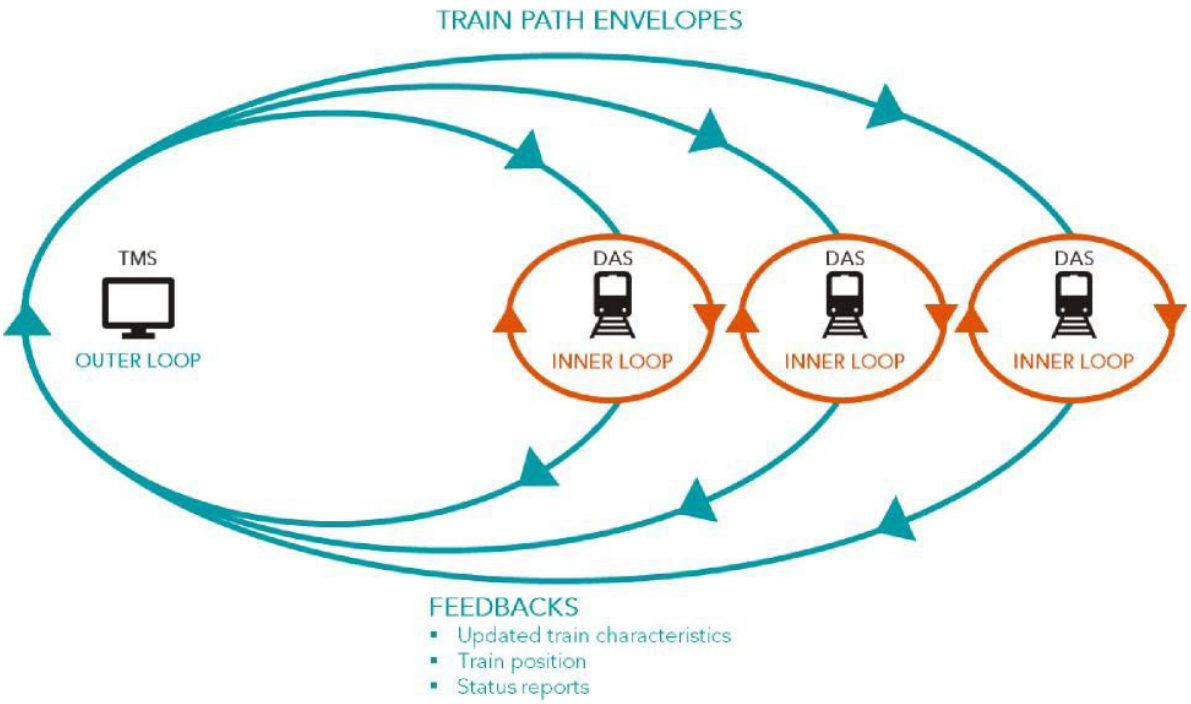


Fig. 8 - DAS inner and outer loops

A DAS only offers advice; train drivers are still expected to make autonomous decisions and drive a train according to the inputs received from the surrounding environment, as well as complying with the legislation and rules of the country in which they are driving, and using the instruments they have at their disposal (including but not limited to DAS).

The main goals of Connected Driver Advisory Systems (C-DAS) are to:

- Improve the smoothness and energy efficiency of a train ride
- Improve the train punctuality
- Avoid timetable conflicts by ensuring trains arrive at the correct point at the correct time
- Reduce maintenance costs by reducing wear and tear thanks to less adverse signals, thus having fewer brake applications
- Optimise the use of capacity and/or increase the regularity of a network by increasing the predictability of train operations
- Improve safety by limiting the number of restrictive signals encountered, and therefore reducing the likelihood of Signals Passed At Danger (SPAD). Using a DAS can also have a positive effect on the driver's stress levels

Additional objectives for DAS should include providing drivers with important information on energy supply (e.g. power limitations) and adhesion, as both factors can consequently influence a journey. The SFERA protocol supports both use cases.

SFERA does not define how the advice should be shown to the driver. Therefore, each RU may determine the design of the interface (DMI), taking national rules and regulations into account.

Examples of DMIs are shown in Fig. 9 - page 45 and Fig. 10.

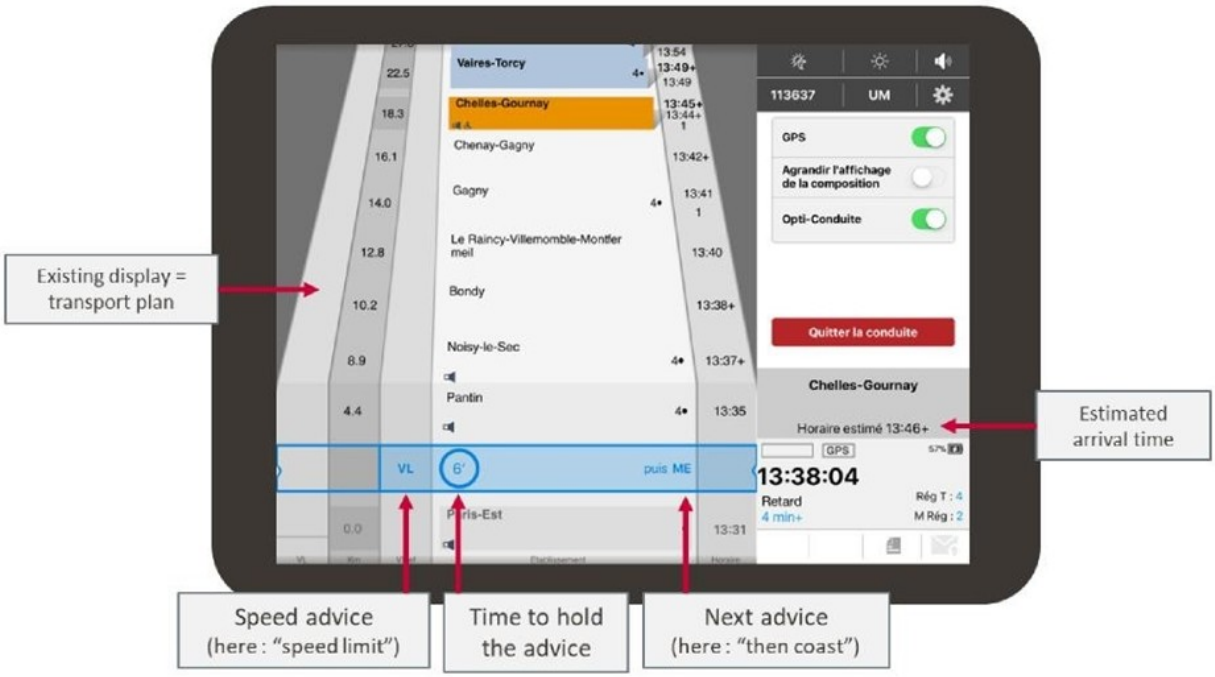


Fig. 9 - Opticonduite, "Standalone DAS" in use at SNCF

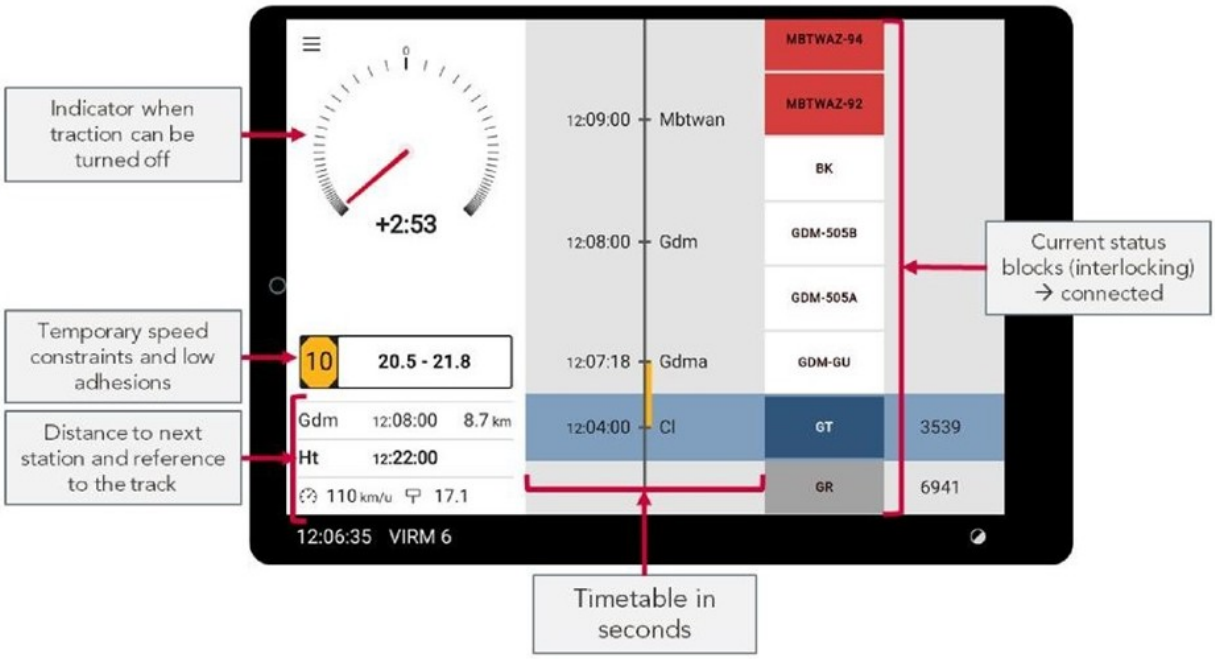


Fig. 10 - TimTim, "Connected DAS" in use at NS

DAS were slowly implemented in Europe over the course of the 2010s, mostly in “standalone” rather than “connected” versions meaning that most systems were not able to exchange data in real time with the TMS used by the IM.

In its first phases of implementation, interoperability was not a key concern, although, now with Connected DAS, which need to be able to switch from one data exchange flow to another as they cross borders, this is becoming ever more important.

3.1.1 – Standalone DAS

A Standalone DAS (S-DAS) has no connection to TMS while driving, but it does have an information source for the position of a train (e.g. GNSS). An S-DAS is pre-loaded with static information describing the train journey, such as:

- Timetable data, the first target of a DAS being to respect TPs

- TCs (type, length, weight, engines, maximum speed, etc.). This data can be updated by the driver with “real” data

- Tracks characteristics (gradients, curves, signals, tunnels etc.). These attributes vary throughout the trip, so the journey will be divided into segments and the characteristics will be defined for each segment

During the trip, the S-DAS generates speed advice for the train drivers according to this pre-loaded data, the position of the train and the current time. The driver can then get information on the position along the journey, notable upcoming points, time differentials (forecasted advance or delay compared to the timetable) and the optimal speed to maintain a smooth, efficient and on-time service.

An S-DAS works by bringing a train as close as possible to the timetable. This is primarily achieved by using existing buffers to absorb smaller delays, thereby preventing escalation into disrupted situations. The S-DAS cannot provide data in a disrupted situation, as it does not receive live information. But it can help to restore a normal situation after a disruption.

Within this IRS, an S-DAS is always considered an S-DAS-O (i.e. with driving advice calculated onboard, see next section). It is not recommended to send DAS-C messages (i.e. driving advice calculated centrally) to S-DAS, as there is a reasonable risk that the DAS will show incoherent advice. Nevertheless, it is possible to use timetable mode with an additional AdvisedSpeed (under TemporaryConstraints in the JP) to give static speed advice when the train is running on time.

3.1.2 – Connected DAS

A Connected DAS (C-DAS) is an evolved S-DAS which is connected to the TMS in order to give and receive updates about a train’s journey. The C-DAS receives updates from the TMS in order to maintain a continuously updated understanding of the factors concerning the journey (e.g. conflicts, new timetables, etc.), and in turn may keep the TMS informed on the status of the train journey.

The DAS continuously recalculates and displays a conflict-free and energy-optimised driving profile from real-time information about a train’s path, infrastructure constraints, topology, timing targets, and so on.

There are three different possibilities within the Connected DAS architecture. The definitions were established by the ON-TIME project[1].

NB: The ON-TIME project uses the term “trajectory”. But for glossary coherence purposes, this IRS uses the term “driving profile”.

The definitions are as follows:

- C-DAS-C (Central): “The [driving profile] and advice are computed centrally. The advice is sent to the On-Board Unit (OBU) for display purposes [only]”.

- C-DAS-I (Intermediate): “A proposed [driving profile] is computed centrally and transmitted to the OBU. In the OBU, the advice is computed and displayed”.

- C-DAS-O (On-Board): “The [computed TPE is transmitted] to the OBU. Here, the optimal [driving profile] is computed, and the advice is calculated according to the [driving profile] and displayed to the driver”.

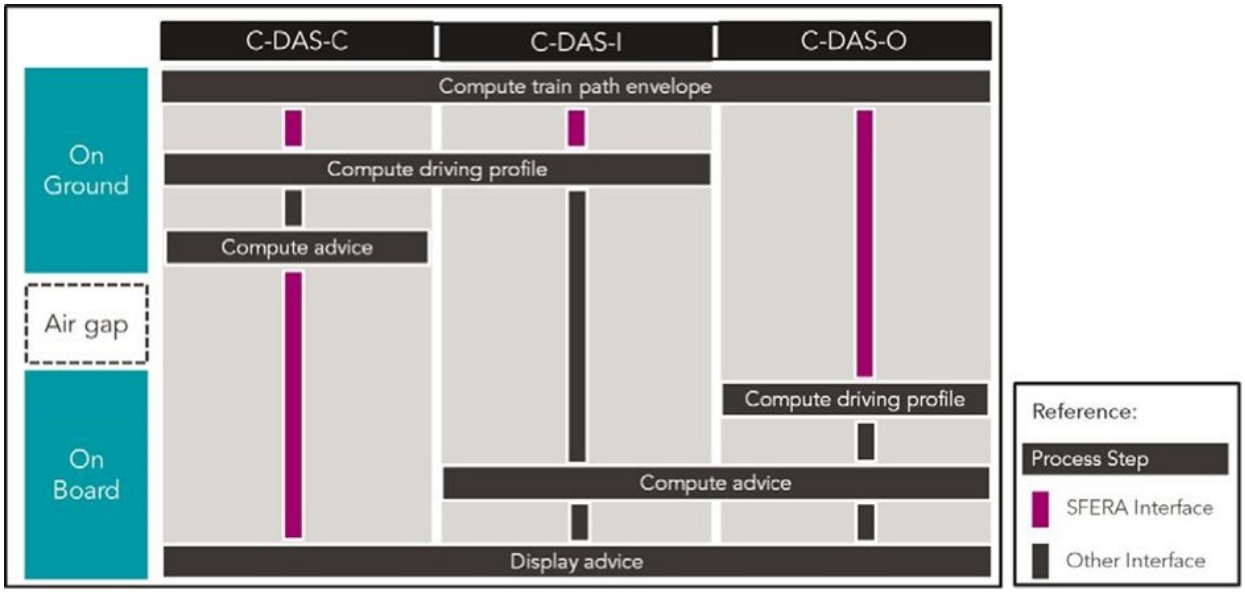


Fig. 11 - Possible C-DAS architectures

The C-DAS architectures are illustrated in Fig. 11. Note that the same three interfaces are present in all three architectures. The only difference is where in the overall process communication between trackside and train takes place. This is often a matter of convenience (reliability and maintainability of the computation and communication devices), price (installation, on-board equipment, etc.) and expected performance (higher or lower latency due to transmission quality depending on the location for the calculation).

This IRS focuses primarily on the **C-DAS-O** approach, as this is also pursued in the ERTMS/ATO model. This choice was made to ensure consistency between the two models. The principal method of C-DAS operation considered here is therefore the transmission of the TPE from the ground to the train, where the driving profile and driving advice are calculated and displayed.

1. https://cordis.europa.eu/project/id/285243.

However, to support railways using C-DAS-C to transition smoothly to C-DAS-O, the SFERA model has been equipped with additional data elements to support **C-DAS-C**. This enables IMs to calculate driving advice locally in the TMS and to send this information to the RUs or directly to the trains for display. This can be used for DAS that are not able to measure position and speed. Only DAS with an architecture setting of “Ground Advice Calculation” (see point 3.1.7 - page 52 for further details about DAS operating modes) may display these advice messages. There are no provisions for transitions between C-DAS-C mode and other modes. No data from C-DAS-C will be displayed in any other mode than C-DAS-C. Localisation for C-DAS-C is provided by the IM, and there are no references to JPs or TCs.

C-DAS-I is not supported in this IRS.

3.1.3 – Automatic Train Operations (ATO)

The functionality of DAS and ATO is very similar: data about the infrastructure, train and timetable is used to calculate a driving profile. The difference between the two systems lies in the execution of this driving profile. The DAS presents it as driving advice to the driver, whereas ATO automatically controls the train’s movements accordingly.

Ongoing projects currently specify data exchange between ground systems and automatic trains on ERTMS lines, so-called ERTMS/ATO. These specifications are defined in UNISIG SUBSETS 125 and 126. SFERA is based on these standards, and all data elements in SUBSET-125/126 are therefore also included in SFERA.

To also enable the use of SFERA on conventional Class B ATP lines, additional data had to be added to the protocol. These mainly concern elements about the infrastructure used for localisation, information about the train itself, and data for object recognition. On ERTMS lines, most of this data is provided by ETCS system components.

This combination of the ERTMS/ATO data model with additional elements ensures DAS functionality. It also allows the SFERA protocol to be used for ATO operations on conventional Class B ATP lines (when permitted by the national safety authority). This approach was deliberately adopted to create a foundation on which SFERA can be upgraded to ERTMS/ATO once a line is fitted with ERTMS.

3.1.4 – The Journey Profile model

To calculate accurate speed advice during a train journey, the on-board DAS needs to be in possession of information about the infrastructure, topology, timetable, and the train itself. In SFERA, this information, sent from the ground to the on-board system, is grouped into three categories:

- Segment Profile (SP): “Static” data of a segment of the infrastructure
- Train Characteristics (TC): The train’s features
- Journey Profile (JP): “Dynamic” data about the train’s journey, especially timings

In return, the SFERA-OB DAS function needs to be able to send information to the trackside concerning the status of the train journey. This is sent through a B2G SR.



Fig. 12 - JP model – information exchange in SFERA

In this IRS, this model of information exchange, shown in Fig. 12, is called a "JP model", and is derived from ERTMS/ATO standardisation (UNISIG SUBSET-125 and 126). The DAS-related SFERA messages have been developed based on SUBSET-125 and 126 to make the SFERA DAS function straightforward for RUs and IMs to use.

However, due to differences in the technical conditions which characterise ETCS and Class B ATP systems, additional data elements had to be included in this IRS to ensure functionality on legacy lines. Furthermore, unlike SUBSET-126, the data exchange in SFERA between ground and on-board systems happens through transmission of XML messages.

SFERA also allows for the transmission of a significant amount of extra data beyond SP, TC, JP and SR. Most notably, SFERA enables the sending of so-called "related train information", which provides data on the preceding and following trains.

3.1.4.1 – Segment profile (SP)

The SP contains mostly static infrastructure data and represents a physical segment of the line, where the infrastructure features are usually not subject to frequent change. This category includes the following data:

- TP locations for which a passing or stopping time may be specified in the JP
- Notable points on the infrastructure (e.g. signals, balises and kilometre reference points etc.) and additional information and constraints regarding these elements (e.g. signal speed restrictions)
- Description of certain areas along the train path (e.g. platforms, tunnels or powerless sections)
- Characteristics of the infrastructure and topology (e.g. maximum speed profiles, curves, gradients, maximum current limitations, etc.)

3.1.4.2 – Train Characteristics (TC)

The TCs contain information about a train's features in order to calculate its dynamic capabilities. As with SPs, the TCs are mostly static since the rolling stock does not change frequently.

While it may seem unnecessary to send data about the train itself from the ground to the train, this is required as a DAS does not necessarily have to be integrated into the cab. It may also be a hand-held device with no direct connection to the train and its technical data. Furthermore, it is essential that ground and on-board systems rely on the same data for calculating running times. The following train features are included, among others:

- Information about the rolling stock type, the number of locomotives and wagons, and traction and braking power

- Physical information about the train, such as its length and weight

- The ATP systems supported by the train

3.1.4.3 – Journey Profile (JP)

The JP is the unifying element that references both the SP and TC. It is the most dynamic of the three data categories. It describes the unique journey (timetable) of a train on the network of a specific IM. In cross-IM operations, the JPs from both data providers shall be merged by the DAS. Among other things, the JP contains the following information:

- Unique train identification for each train (see point 5.5 - page 69 for more details)

- Timing restrictions that define the limitations within which the train should operate, including time tolerances and stopping point information

- Dynamic infrastructure data such as temporary constraints (e.g. low adhesion conditions or temporary speed restrictions) and dynamic information related to the ATP system with which a line is equipped

- References to TP locations defined in the SPs and references to the TCs

3.1.4.4 – Status report (SR)

An SR is the means by which the SFERA-OB/DAS unit sends information to the SFERA-TS/DAS concerning the position of the train, or data related to its current condition. For instance, the SR may be used to send the following information:

- The position and speed of the train

- The most recent TPs passed or a forecast of when the next TPs will be reached

- The JP currently in use

- Changes in the characteristics of the train

- Any changes in adhesion conditions

- Data on expected current consumption and regeneration over the next few minutes

- The status of the train service: train ready/not ready at origin, interruption of the train run and the description of delay causes

3.1.4.5 – Related train information

The related train information provides a means of exchanging data between trackside and train regarding line occupation and the status of other trains operating on the same track.

This makes it possible to share the position and speed of these trains, for example. A DAS may use this information to update advice, or drivers may use their expertise to interpret this. Additionally, a driver can inform passengers of potential delays caused by other trains affecting the timetable.

The related train information can also improve the acceptance of DAS by drivers. It serves as contextual information in cases where advice appears to contradict the planned timetable.

3.1.5 – Application

In an S-DAS setup, all data is pre-loaded on the DAS device before the train's departure and remains unchanged throughout the journey. Additionally, no SR messages are possible. In contrast, a C-DAS is frequently updated when there are changes, such as temporary speed restrictions, path or timing alterations, etc., and also supports the B2G transmission of SRs.

In the data sent to the SFERA-OB/DAS, the JP is the unifying element that references both SPs and TCs, which may be multiple. Each SP and TC instance has an ID and a version number, which the JP references. A train journey may consist of multiple JPs, for example, when a train's designation changes during the journey or when a train crosses a border into another IM's jurisdiction.

Generally, JPs change more frequently than SPs and TCs. This is because timetable information is dependent on operational conditions, whereas a train's path and rolling stock typically remain fixed during a journey. When a train path or rolling stock changes (e.g. due to re-routing or when units are coupled), a new set of data is created with an updated ID number. If the DAS encounters a new SP or TC ID number or version in a JP, it will request the updated data from the TMS and recalculate the driving advice.

To limit transmission size, the more dynamic JP contains only a small amount of data. The larger SP and TC data sets can be pre-loaded and stored locally on the SFERA-OB/DAS.



Fig. 13 - Data architecture of a sample train journey

The example given in Fig. 13 shows the data architecture of a train journey with two JPs and multiple SPs and TCs.

3.1.6 – Additions to ERTMS/ATO

The *IRS 90940* model aligns as closely as possible with the ERTMS/ATO standard (UNISIG SUBSET-125/126). This was intended to minimise the effort required by IMs to generate *IRS 90940* data in parallel with UNISIG ATO data and to offer similar data for DAS and ATO to RUs. This approach also ensures that *IRS 90940* data contains all of the information necessary for ATO operations.

However, unlike ERTMS/ATO, *IRS 90940* aims to support Class B ATP lines. Due to the technical constraints of these legacy lines (e.g. the absence of ETCS balises or ETCS On-Board Units), this

IRS includes additional data elements which would typically be supplied by ETCS on ERTMS lines. These especially comprise:

- The TCs: Those used by ERTMS/ATO are either received from the TCMS, manually entered by the driver during the data-entry procedure or hardcoded in the ATO via the ETCS On-Board Unit.

- Additional infrastructure information: This is required by the DAS to determine a train's location (with ETCS, this is provided by balises).

- Additional signalling system information: For example, the position of signals, typical signal aspects and the signal-induced driver behaviour (the actual signal aspect is unknown at the time the driving advice is calculated).

- Context information about the train journey: Helps the driver interpret the advice given (e.g. kilometre reference points, reasons for timetable updates, etc.).

Furthermore, while SFERA messages are exchanged in XML format, SUBSET-126 uses a binary format.

A detailed table for the conversion of data elements between the two models can be found in Appendix G - page 89.

SFERA also includes reference data to assist companies in associating the exchanged data with the *TAF/TAP TSI protocol*. This includes the TAF/TAP train identifiers, and the locations used in TAF/TAP (PLC/SLC identifiers).

3.1.7 – DAS operating modes

As SFERA supports various setups (see point 1.3.1 - page 25), it is essential that ground systems are informed of the operating modes available on the SFERA-OB/DAS to ensure correct interaction.

For instance, the on-board system may operate in Standalone DAS mode, may only support C-DAS-C, or, in the case of multiple hand-held devices (e.g. for training purposes), only one should be considered held by the train driver (and therefore permitted to send data from board to ground), while the others should be "Read-Only", i.e. only able to receive the G2B messages.

A DAS operating mode consists of three different components:

- Driving mode (DM)

- Architecture

- Connectivity

When establishing a session, the SFERA-OB/DAS sends all supported triplets to the SFERA-TS/DAS. The SFERA-TS/DAS replies to the SFERA-OB/DAS with the pair [architecture, connectivity] selected.

Driving mode is an operational choice and may change during the train's journey. In specific cases, a SFERA-OB/DAS may be "forced" to change its driving mode. The state changes between different modes are described in the DM use cases, see Appendix A - page 83.

If either the SFERA-TS/DAS or the SFERA-OB/DAS needs to change architecture and/or connectivity, the session shall be terminated, and a new handshake initiated.

3.1.7.1 – Driving mode

The DAS Driving Mode of an on-board device may be:

- *Inactive*: The device can receive JPs but is not currently providing information to the user. This may apply to an S-DAS downloading data for a future train service, or a C-DAS updating data in the background, but not presenting advice to the driver.

- *Read-Only*: The device is operated by someone who is not necessarily the train driver. This may be due to RU limitations or because another person (e.g. a future driver, trainer, or debugger) is using the device. All advice messages are available, but position, speed, and power data shall not be sent in the B2G SR. Certain SFERA-TS/DAS implementations may allow either a single connected device (i.e. with Read-Only devices being either prohibited or permitted by the IM, e.g. GoA1 or GoA2) or multiple Read-Only devices.

- *Timetable*: The device is operated by the train driver. No advice is presented to the driver, only the objectives. This may be due to the IM not giving sufficient information to calculate the advice on-board (no TMS present, or insufficient data for SP generation), or the RU not implementing a DAS-O algorithm.

- *DAS not connected to ATP*: The device is operated by the train driver. The advice calculated by the DAS may conflict with safety orders from the signalling system.

- *GoA1*: The device is operated by the train driver. The device is connected to the active ATP system, ensuring advice does not contradict ATP safety conditions.

- *GoA2/3/4*: SFERA data is used to operate a train under ATO GoA 2/3/4. Driving profiles are used for direct train control. SRs include reliable data on TCs, adhesion and power forecasts from the TCMS. Text messages and C-DAS-C advice messages are presumed ignored as there is no automated way to interpret them.

There are common elements for the “DAS not connected to ATP”, “Timetable” and “GoAx” driving modes:

- The device is operated by the train driver. The RU shall ensure that this is the operational driver. The SFERA-TS/DAS shall ensure that only one device is in any of the driving modes at any given time.

- The train position is measured at the front of the train (default) or from another point of the train. In the latter case, the distance between this measurement point and the front of the train must also be provided in the report. Data updates sent to this device will be visible to the driver.

3.1.7.2 – Architecture

Architecture may be:

- *Ground Advice Calculation* (i.e. DAS-C): Advice is calculated on ground systems and only speed advice is sent to the SFERA-OB/DAS.

- *Board Advice Calculation* (i.e. DAS-O): The SFERA-TS/DAS produces and sends TPEs to the SFERA-OB/DAS, which calculates and displays appropriate speed advice.

3.1.7.3 – Connectivity

Connectivity may be:

- *Standalone*: The DAS uses a previously downloaded JP. It may or may not process JP updates. SRs are sent, based on the original JPs. In IM-RU setups, the RU back office may send data directly to the IM back office, if cached versions are sent with high latency to the on-board DAS.

- *Connected*: The DAS will respond quickly to JPs being updated. Therefore, the trackside can assume that the most recent one is in use. SRs will be sent, stating the version of the JP on which the current advice to the driver is based.

3.2 – Digital Instructions (DI)

3.2.1 – Introduction and context

In the context of train traffic management and real time operations, several situations may arise which currently are not addressed by signalling systems, whether ETCS or Class B. To maintain an acceptable level of safe operations at all times, *Appendix C2* of the *OPE TSI* defines a series of harmonised European Instructions and the concept of National Instructions, collectively referred to as Operational Instructions. These instructions are intended to provide clear and unambiguous authorisations or impose additional restrictions beyond those of a signalling system. The communication of these instructions is a vital process within the envisioned safety environment.

Despite various security measures, a residual safety risk remains due to potential misunderstandings in verbal communication, particularly when the operating language is not the driver's mother tongue. Additionally, some IMs still rely on the physical delivery of instructions to mitigate verbal communication errors, which requires robust logistical and administrative processes.

Considering the challenges in communicating operational instructions, as mentioned above, several IMs have started developing digital solutions to the problem. To harmonise these and future efforts, the SFERA data exchange protocol incorporates a Digital Instructions (DI) function.

3.2.2 – Description of an instruction

3.2.2.1 – The format of a SFERA Digital Instruction

The format of the SFERA compliant instructions is standardised to include:

- **A header**: The leading section of an instruction. If multiple instructions (bodies) are grouped in one payload, they shall all share the same and single header.

- **A body**: Consisting of one or more sections that contain the actual instructions.

- **A footer**: The closing section of an instruction. If multiple instructions (bodies) are grouped in one payload, they shall all share the same and single footer.

Within the header, body, and footer, **sections** can exist. These are defined to group fields and provide a coherent structure. This enhances the readability and performance of the XML and XSLT code.

Each section contains dedicated elements to carry information:

- **Label**: Official information regarding a tick box or a field.
- **Tick box**: Ability to tick or untick part of an operational instruction. When ticked, the corresponding label is active.
- **Field**: Input options corresponding to parts of the operational instruction. When filled in, the corresponding label is active.

3.2.2.2 – European Instructions

European Instructions are defined in *Appendix C2* of the *OPE TSI*, each serving a specific purpose related to train movement and safety.

Their application is defined in Appendix A of the *OPE TSI*, and shall not be altered by the data exchange protocol. Nevertheless, 3 types of European Instruction can be distinguished by title:

- Authorisation
- Obligation
- Revocation

The revocation type is specifically used to withdraw an instruction, to indicate that the receiver should no longer take it into account.

Appendix C2 of the *OPE TSI* provides a general visual overview of each European Instruction, specifying that every tick box, field, and input option within an instruction must have a unique alphabetical or numerical identifier, with shared identifiers having the prefix ‘x’, unless digitally transmitted. While the content and order of these identifiers must be preserved, the format of the visualisation is indicative.

For example, IMs and RUs may choose not to display unused tick boxes or fields for their network but are prohibited from adding new ones. IMs and RUs may also add guidance on completing and interpreting the forms, provided that it remains separate from the official communication procedure.

Building on these principles, the following prescriptions apply to the SFERA protocol:

- Any added guidance shall not be included in the exchanged data.
- The visualisation of an instruction shall, at a minimum, include all relevant tick boxes (checked) and fields (filled) present in the payload of the SFERA message, accompanied by their respective labels.
- XSLT shall be used to visualise exchanged XML data in the business tool user interface. The IM shall provide XSLT files for every instruction used on its network. The RU may choose to use, adapt, or create their own XSLT files.

For XSLT, see Appendix I - page 91.

3.2.2.3 – National Instructions

The OPE TSI gives IMs the freedom to compose National Instructions for safety procedures or processes not covered by European Instructions. The format and application of these instructions are IM-dependent and have not necessarily been harmonised.

As for the SFERA protocol, the following prescriptions shall be taken into account:

- Any added guidance shall not be part of the exchanged data.

- The visualisation of an instruction shall, at a minimum, include all relevant tick boxes (checked) and fields (filled) present in the payload of the SFERA message, accompanied by their respective labels.

- XSLT shall be used to visualise exchanged XML data in the business tool user interface. The IM shall provide XSLT files for every instruction used on its network. The RU may choose to use, adapt, or create their own XSLT files.

For XSLT, see Appendix I - page 91.

3.2.3 – Instruction lifecycles

Without intending to change the operational regulations, the “life” of an instruction, from creation to archiving, is given as the “Instructions lifecycle” within the SFERA protocol. The purpose of this lifecycle is to harmonise key phases and statuses so that, at any given time and especially in case of a malfunction, each party can clearly indicate the extent to which the instruction is valid or not.

3.2.3.1 – Lifecycle phases

Each phase represents a conceptual timeframe that comprises certain subprocesses and may result in a status change. They are used to distinguish intermediate steps in the communication of operational instructions.

Table 8 outlines the different phases of the lifecycle and their respective purpose. A diagram is included to represent the flow through these phases in Appendix A - page 83 A of this IRS.

Table 8: The Instructions Lifecycle Phases

Phase	Description	Status
Conception	The instruction is developed and prepared. During this phase, information from the receiver may be requested, such as their current location. The instruction remains adaptable and cannot be published in this state.	Concept
Finalisation	After the operational instruction has been created, the sender performs a verification. During this phase, the instruction cannot be edited, although a return to the conception phase is possible if needed. Once verified, the instruction is marked as "finalised" and is ready to be published.	Finalised
Publication	The sender's tool initiates the sending process and starts the publication phase. The instruction shall, at a minimum, be in a "finalised" state. Upon publication, the instruction is timestamped, assigned a unique identifier, and made available on the IM SFERA-TS.	Published
Retrieval	This phase comprises the request of the instruction by the receiver, its sending by the sender system, and its verification by the receiver. The request step shall be omitted where the two-way verification method is not applied. The instruction is marked as 'retrieved' once the preconditions for sending it are satisfied, e.g. a successful request or the omission of the two-way verification method.	Retrieved
Confirmation	In this phase, the receiver confirms verification and compliance with the instruction. In connectivity mode "full", this process is digitalised or it is carried out via verbal communication in connectivity mode "limited" (see point 3.2.4.2).	Confirmed
Execution	This phase spans from confirmation to completion. During execution, a findings report may be filed, and the instruction can be declared as "Executed".	Executed
Completion	The completion phase is the final phase of the lifecycle. Completion constitutes the irreversible action of ending the instruction's lifecycle. This action is the responsibility of the receiver, except in specific degraded situations (see Appendix A)	Completed

A Digital Instruction is considered issued in accordance with *TSI OPE* once the receiver confirms compliance, whether this confirmation is given digitally or verbally.

The **archiving** of an instruction occurs after it reaches its final state: "completed", "cancelled" or "revoked" (see point 3.2.3.2). Archiving is carried out independently by the IM and RU. If needed, bilateral archiving agreements may be established between the IM and RU, but these must be arranged outside the scope of the SFERA protocol.

3.2.3.2 – Instruction statuses

While an instruction is in a given state, certain actions can be performed while others cannot. These statuses are essential for providing a clear overview of the instruction at distinct points in its lifecycle.

Table 9 outlines the possible SFERA statuses and their implications.

Table 9: Instructions statuses

Status	Description and implications	Preceded by	Followed by
Concept	After creation, and during conception, it is recommended to label the instruction as a “Concept”. While in this status, the instruction’s contents can be edited.	None	Finalised; cancelled
Finalised	Following verification by the sender, the concept becomes “finalised”. A “finalised” instruction’s body shall no longer be editable; however, the header and the footer sections may still be modified. A “finalised” instruction is ready to be published but is not yet retrievable by the receiver.	Concept	Published; cancelled
Published	Once the publication phase is successfully executed, the instruction is marked as “published”, meaning it can now be retrieved from the IM SFERA-TS, but only for devices in the “DI Active Driver” driving mode.	Finalised	Retrieved; cancelled
Retrieved	A “retrieved” instruction is one for which all preconditions for sending are met. This status follows publication and indicates that the instruction can be pushed to the receiver for verification. Once retrieved, the instruction’s lifecycle can no longer be cancelled but shall, if necessary, be revoked in accordance with the prescriptions of the *TSI OPE*.	Published	Confirmed; revoked
Confirmed	Indicates that the receiver has verified and confirmed the instruction, and will comply with its content. Note: As soon as an instruction reaches this status, it will be considered as **issued** according to the *OPE TSI*.	Retrieved; executed	Executed; completed; revoked
Executed	The “confirmed” instruction is marked as “executed” in the receiver’s application, allowing business tools to adapt their visualisation or responses accordingly. Note 1: European Instruction No.3 cannot be set as “executed” (see *OPE TSI*). Note 2: European Instruction No.4 cannot be set as “executed” as it is automatically set as “completed” during revocation.	Confirmed	Completed; confirmed

Table 9: Instructions statuses

Completed	Marks the irreversible end of an instruction's lifecycle. The instruction can be "completed" if it has previously been "executed". Note 1: European Instruction No.3 cannot be set as "completed" (see OPE TSI). Note 2: European Instruction No.4 is automatically set as "completed" during revocation.	Executed	None
Cancelled	Irreversibly aborts the lifecycle of an instruction.	Concept; finalised; published	None
Revoked	Irreversibly aborts the lifecycle of an instruction.	Retrieved; confirmed	None

3.2.3.3 – Processes During an Instruction's Lifecycle

Throughout each phase of the lifecycle, and in order to transition between statuses, several subprocesses are defined within the protocol. These are described in more detail in Appendix A - page 83.

3.2.4 – DI operating modes

Since a fully digital DI exchange is only possible if all of the system's components are aligned and/or functioning, it is essential to distinguish between different operating modes.

For instance, a Digital Instruction may start its lifecycle in a fully digital data exchange environment. However, due to connectivity issues, the connection between the trackside and on-board system may not always be restorable. Despite this, the signaller and train driver still need to conclude the regulatory procedures for the instruction at hand. The TS and OB will continue to function in a degraded connectivity state with adapted functionalities. Therefore, SFERA/DI operating modes are designed to retain certain functionalities even in less or unconnected states.

Another example is when crossing borders. If the IM-RU setup is no longer applicable, an IM-Train setup may need to be considered. In such scenarios, the establishment of a new DI Operating Mode ensures the continuity of operations and the handling of instructions.

A DI operating mode consists of two different components:

- Driving mode (DM)

- Connectivity

When establishing a session, the SFERA-OB/DI sends all supported pairs to the SFERA-TS/DI. The SFERA-TS/DI replies to the SFERA-OB/DI with the supported connectivity mode.

Driving Mode is an operational choice and may change during the train's journey. In specific cases, the SFERA-OB/DI may be "forced" to change its driving mode. The state changes between different modes are described in the Data Exchange (DX) use cases, see Appendix A - page 83.

If either the SFERA-TS/DI or the SFERA-OB/DI needs to change connectivity, the session shall be terminated and, whenever possible, a new handshake initiated.

3.2.4.1 – Driving mode

The DI driving mode of an on-board device can be one of the following:

- *Inactive*: The device can receive instructions and status updates but is not currently providing information to the user.

- *Read-Only*: The device is operated by someone who is not necessarily the train driver. This may be because the user has not yet been authenticated as the active driver or because another person (e.g. a future driver, trainer, or debugger) is using the device. All instructions from "retrieved" (included) to "completed" (excluded) are available on the Read-Only device. Status updates sent from B2G by a Read-Only device shall not be taken into account by the SFERA-TS/DI.

- *DI Active Driver*: The device is operated by the train driver. The DI Active Driver is the only device which can interact with the SFERA-TS regarding DI subprocesses and DI status changes. The SFERA-TS shall ensure that only one device is a DI Active Driver at any given time.

3.2.4.2 – Connectivity mode

Table 10 illustrates the different DI function connectivity modes regarding operation of SFERA-TS/DI, SFERA-OB/DI and the SFERA data exchange connection.

Table 10: Connectivity modes for the DI function

ID	Connectivity Mode	SFERA-TS/DI	SFERA Data Exchange	SFERA-OB/DI	DI Status Updates
4	Full	Yes	Yes	Yes	Digitally exchanged
3	Limited	Yes	Yes	Yes	Confirmation of the instruction by the driver is given verbally and the status "confirmed" is recorded by the signaller
2	Offline	Yes	No	Yes	Status updates are made in the signaller's tool and are available on the SFERA-TS
1A	Offline	Yes	No	No	Status updates are made in the signaller's tool and are available on the SFERA-TS
1B	Offline	No	No	Yes	Status updates are made in the on-board application and are available on the RU SFERA-TS

It is important to note the following connectivity-specific prescriptions:

- In this version of the SFERA protocol, during the lifecycle of an instruction, it is not currently possible to upgrade to a higher connectivity mode. This means that if the connectivity has been degraded (e.g., due to technical issues), the instruction shall be managed in this connectivity, or a lower one, until completion.

- For “offline” connectivity modes, it is impossible for an instruction to have the status “retrieved”, due to the absence of digital transmission.

- The choice between the use of a “full” or “limited” connectivity mode is made by the IM and shall be communicated during the handshake process.

3.3 – Free Text Communication (FTC)

3.3.1 – Introduction and context

Free Text Communication (FTC) is a protocol to be used to exchange information between the driver and the ground systems/staff in situations where:

- The complex procedure for communicating Digital Instructions is not necessary, given the safety level of the information to be communicated.

- The nature of the information to be exchanged is not sufficiently common in railway operations to be included in the structured data model of the C-DAS function.

3.3.2 – Functionalities

The purpose of FTC is to exchange plain text messages between the driver and the ground. The following functionalities are provided:

- Sending a message from the driver to a ground contact

- Sending a message from a ground contact to the train driver

- Conversations: the recipient of a plain text message can refer to its identifier in a response, allowing the original sender to see an explicit link in their application

- Sending multiple versions of a plain text message in different languages

- Sending all messages exchanged for a given train to third party devices

- Directory: the SPs (payload shared with DAS) can include a list of ground contacts who can receive plain text messages, so that the driver can explicitly choose the recipient

The SFERA protocol is not designed for conversations between more than two parties. The messages are exchanged solely between a train driver and a unique ground identifier. The IM is responsible for defining the identifiers, and who they refer to:

- If a different identifier is used for each person on the ground, conversations can only occur between the driver and that identifier. However, certain implementations may present a Read-Only view of the conversation to others for synchronisation purposes, if needed.

- If a unique identifier is used for multiple people, all of them can add messages to the conversation. The aggregation of these messages shall be handled within the business tools, before being sent through the SFERA-TS/FTC.

If a ground contact needs to broadcast a unique plain text message to all the drivers within a certain zone, it shall be duplicated and sent via separate payloads addressed to each individual train being targeted.

3.3.3 – Driving mode

The FTC driving mode of an on-board device can be one of the following:

- *Inactive*: The device can receive plain text messages exchanged between the active driver and ground contact but is not currently providing information to the user.

- *Read-Only*: The device is operated by someone who is not necessarily the train driver. This may be because a non-driver (e.g. a future driver, trainer, or debugger) is using the device. All plain text messages exchanged for a given train are sent to Read-Only devices. Plain text messages sent from B2G by a Read-Only device shall not be taken into account by the SFERA-TS/DI.

- *FTC Active Driver*: The device is operated by the train driver. The FTC Active Driver is the only device which can send plain text messages to the SFERA-TS. The SFERA-TS handling devices (the RU SFERA-TS in an IM-RU setup, and the IM SFERA-TS in an IM-Train setup) shall ensure that only one device is an FTC Active Driver at any given time.

4 – Use cases and prerequisites

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

4.1 – General presentation of the use cases

The data structure and communication layers of SFERA have been defined by taking a set of use cases into account. These represent the interaction between a SFERA-OB and the signaller and traffic manager tools on the ground, as well as external entities (e.g. other trains or power supply management applications), and are available for consultation in Appendix A - page 83.

Some of the use cases described operate at a process level, while others are at data exchange level. The use cases may be interpreted differently based on the setups used (see point 1.3.1 - page 25 for a description of the setups).

- For process level use cases, the stakeholders listed refer to the applications used to perform actions on the data (e.g. TMS, DAS, RU Backoffice, traction power management systems, etc.). These use cases are directly applicable, no matter if the IM-RU or IM-Train setup is used;

- For exchange level use cases, the systems communicating with each other differ for each setup. These refer to a communication between a receiver and a provider. The "receiver" is the system nearest to the business tool on-board, and the "provider" is the system nearest to the signaller's tool.

Table 11 summarises the roles of the different systems in the three types of links that may be encountered.

Table 11: Provider and receiver roles in different setups

Setup	Link	Provider	Receiver
IM-RU	IM SFERA-TS <-> RU SFERA-TS	IM SFERA-TS	RU SFERA-TS
	RU SFERA-TS <-> SFERA-OB	RU SFERA-TS	SFERA-OB
IM-Train	IM SFERA-TS <-> SFERA-OB	IM SFERA-TS	SFERA-OB

In addition, sequence diagrams are described to provide examples of use case implementation using SFERA messages. For the process level use cases, the sequence diagrams are described in the context of the IM-Train setup, which uses a push configuration. The use case appendix also includes an example of how the sequence diagrams can be extended for the IM-RU setup.

In addition to the use cases, a selection of real-life scenarios (referred to as implementation cases) is available in Appendix C - page 85, which presents:

- A range of use cases,

- The events that occur in an IM-Train setup at communication level (see point 5 - page 66).

Consulting this appendix may provide additional insight into the links between the use cases and interactions with the communication layer for the IM-Train setup.

4.2 – Prerequisites

Prior to operational implementation of the use cases described in Appendix A - page 83, certain prerequisites shall be agreed upon. This step must be repeated whenever an RU begins using a particular SFERA function on a new IM's network, or following a major systems upgrade. This step is not governed by a specific protocol, and therefore this IRS does not prescribe a method. However, the following conditions are required to be met before each next step can be taken:

1. The RU knows how to contact the SFERA server of the IM

2. The IM and RU have agreed on the time synchronisation accuracy

3. The RU has sufficient knowledge on the service level of the IM SFERA server

4. The IM and RU agree that the use of the specific SFERA function will not negatively impact operations on the IM's network, in accordance with the processes guiding the implementation of change (e.g. Common Safety Methods)

5. The RU has implemented a process for applications to obtain the correct Train-IDs and identifiers relevant to each IM area

6. Optional: There is a Service Level Agreement for the SFERA data

7. Optional: There is an agreement between the IM and RU on the minimum bandwidth requirements

For the IM-Train setup only:

8. The IM knows the validation mechanism for the authorisation protocol and all related data from the RU

9. The IM and RU have confirmed their commitment to the SFERA requirements

For the "IM-RU" setup only:

10. The IM and RU have shared the respective common interface instance numbers to be used for SFERA data exchange

For the DAS function only:

11. The IM and RU have agreed on the mode(s) for which data is available on the IM's network (C-DAS-O and/or C-DAS-C)

12. The IM knows if SRs are possible from the RU's DAS

13. The IM and RU have implemented the processes to ensure the IM has access to TCs information

14. The RU has implemented a process for the DAS to identify the SFERA-TS to connect to in order to obtain the JP for the initial part of the journey where the DAS will be used

For the DI function only:

15. The IM shares XSLT files for all instructions potentially exchanged on its network, enabling the RU to process them consistently or adapt them to its operational needs

16. It shall be ensured that the messages and notifications are being exchanged between the signaller and the correct train driver.

Outputs [1, 2, 8, 9, 10] are used in the communication protocol, and [3, 4, 5, 6, 7, 11, 12, 13, 14, 15, 16] to ensure effective operation.

5 – Data Structure

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

The present point provides an overview of the SFERA data structure, i.e. the actual content of the messages exchanged between on-board devices and the ground.

The complete description of the XML message format which can be sent in SFERA is provided in an XML Schema Definition (XSD) available in Appendix D - page 86. This is complemented by a "Data Handbook" available in Appendix E - page 87, which gives detailed information on the SFERA data structure, as well as guidelines and implementation suggestions. Appendix E also provides details on the connection modes between DAS subsystems.

Appendix F - page 88 contains sample XML communications that can be used as a foundation for building workable SFERA XML files.

5.1 – Messaging

The SFERA protocol exchanges data in XML format between on-board and trackside systems. This is done through B2G and G2B messages.

G2B messages generally go from the IM SFERA-TS to the SFERA-OB, and B2G messages are sent in the opposite direction.



Fig. 14 - G2B vs. B2G in IM-Train Setup



Fig. 15 - G2B vs. B2G in IM-RU Setup

Therefore, between an IM SFERA-TS and an RU SFERA-TS, G2B and B2G messages are exchanged, although both functions are technically located on the on trackside.

For B2G and G2B, three types of message can be generated:

- Event messages:
  These are unrequested messages used to communicate events.

- Request messages:
  These are messages used to request data.

- Reply messages:
  These messages are sent as a reply to either event or request messages.

Therefore, there are six types of SFERA messages in total:

- B2G Event/Request/Reply

- G2B Event/Request/Reply

All SFERA messages include a message header and a payload. The structure of the message header is the same for all types of messages, while the payload varies according to the type of message.

A set of six of these types of SFERA messages are defined for every function supported by the protocol (DAS, DI, and FTC).

5.2 – Message headers

Each SFERA message has a message header, which contains several fields. This includes, for instance, the version of the SFERA protocol used by the message, the timestamp, the message source and destination identifiers, as well for of the message, as well as other optional data.

Each SFERA message header shall contain a message ID, which is assumed to be unique.

When a SFERA message needs to refer to earlier communications (e.g. SFERA reply messages and C-DAS-C revoke messages), the so-called "correlation ID" is used. This field can be populated with the ID of the previous message to which the current message refers.

The SFERA message headers include elements shared with the TAF/TAP message header. This is necessary to enable the transmission of standard SFERA messages via Common Interfaces, used as a communication layer for the IM SFERA-TS <-> RU SFERA-TS link. Appendix B - page 84 provides implementation details of the message header regarding the Common Interface.

5.3 – Handshakes, session termination, and driving modes

Once the communication layer is established between two SFERA entities, two application-level transactions – the handshake, and the driving mode – need to be carried out for each SFERA function, based on the level of communication exchange needed for a train.

The handshake and driving mode transactions of the various SFERA functions are carried out through separate messages to allow the different functions to be implemented independently. However, common principles and message sequences are applied to facilitate implementation:

- The handshake is required to send business information from board to ground for a specific train. It is initiated by the board side of the communication, resulting in an agreed set of parameters and functionalities being defined for use during the application session.

- The driving mode determines whether a given device is permitted to send business information to the ground. Once the handshake has been established, the SFERA-OB is, by default, considered to be in “Read-Only” mode. The driving mode is upgraded to an “active” driving mode (e.g.: “DI Active Driver”) through dedicated transactions. The SFERA-TS is responsible for ensuring that only one SFERA-OB function has an “active” driving mode at any given time.

Session termination transactions are also supported in the SFERA protocol, allowing the application session previously established through the handshake to be ended. As with the handshake and driving mode transactions, common principles and message sequences are used. A session termination means that a device’s driving mode is no longer valid.

5.4 – Message payloads

SFERA message payloads vary according to the type of message. The main uses of the messages are:

- B2G event messages are used to send SRs (as described in point 3.1.4.4 - page 50), DI status updates, and plain text messages.

- B2G request messages are used to send data requests to the ground. This is primarily used to ask for SPs, TCs, JPs, and DIs (see point 3.1.4.1 to point 3.1.4.3 - page 50), but some other types of data may also be requested.

- B2G reply messages are used to send acknowledgements (OK or errors) to ground messages, but they can also be used to send information about the driving mode (see point 3.1.7 - page 52).

- G2B event messages are used to communicate updates to the JP, the status change of a DI, or send a plain text message.

- G2B request messages are used to request the status of the SFERA-OB.

- G2B reply messages are used to respond to B2G event or request messages. They either contain current data (e.g. the JPs, SPs, TCs or DIs requested by the SFERA-OB), or responses to the messages received, which can be positive acknowledgements or errors.

SFERA message payloads can contain a wide range of data that provides information to be exchanged between on-board and ground systems. For an in-depth description of the data, please refer to the XSD in Appendix D - page 86 and to the Data Handbook in Appendix E - page 87.

NB: TAF/TAPTSI reference data can be included in the SFERA messages to help implementations that link SFERA data with TAF/TAP data. The TAF/TAP locations are used in TAF/TAP messages to link elements to the infrastructure. For example, the planned date/time of a train’s arrival into a station is linked to the TAF/TAP train location. The list of

TAF/TAP locations for European IMs is available in the Common Reference Database (CRD), accessible through web services or file extractions

5.5 – Identifying JPs, TCs and SPs

SPs and TCs have both identifiers (IDs) and versions. These IDs and versions are referenced in JPs to create a train route. If a JP sent to the SFERA-OB references SPs and/or TCs with IDs and/or versions that are not already loaded onto the SFERA-OB, the On-Board Unit requests these “new” items from the ground.

In contrast, JPs do not have IDs. A JP is uniquely associated with a train via the “train identification”, which is used in several SFERA messages. In particular, it links a JP to a specific train.

In SFERA, a TAF/TAP identifier (following TAF/TAP specifications) or an Operational Train Number (OTN) identifier may be used. While a TAF/TAP identifier is preferable, for transition purposes, OTN identifiers are also supported by SFERA.

Although JPs do not have IDs, they do have versions. While the assumption is that the most recently received JP is always correct, with the timestamp of the SFERA header ensuring that updates are not processed in the wrong order, the JP version is used to check whether the JP in use by the SFERA-OB matches the latest version sent by the trackside unit.

The TMS can therefore use the JP version as a unique identifier. It is also used by the DAS to send an SR to the trackside, communicating which JP is currently being used to generate advice. Using the SR, where possible, the DAS will also report the train’s position within a segment. These SR messages allow the TMS to validate that the DAS is operating on the most recent information.

5.6 – Error codes

As mentioned above, SFERA reply messages (both B2G and G2B) can contain acknowledgements to event and request messages. These acknowledgements may be positive (simple “OK” messages) or can contain an error.

When sending an error, the reply message can identify the location (XML Path) of the original message and provide an error code to specify the reason for the error.

A list of SFERA error codes is available in Appendix H - page 90.

5.7 – Message XSLT transformation

The SFERA XML2HTML is a DI processor implemented entirely in XSLT. It transforms a SFERA DI schema document into a HTML document that can be rendered by any HTML5 compatible client application or browser.

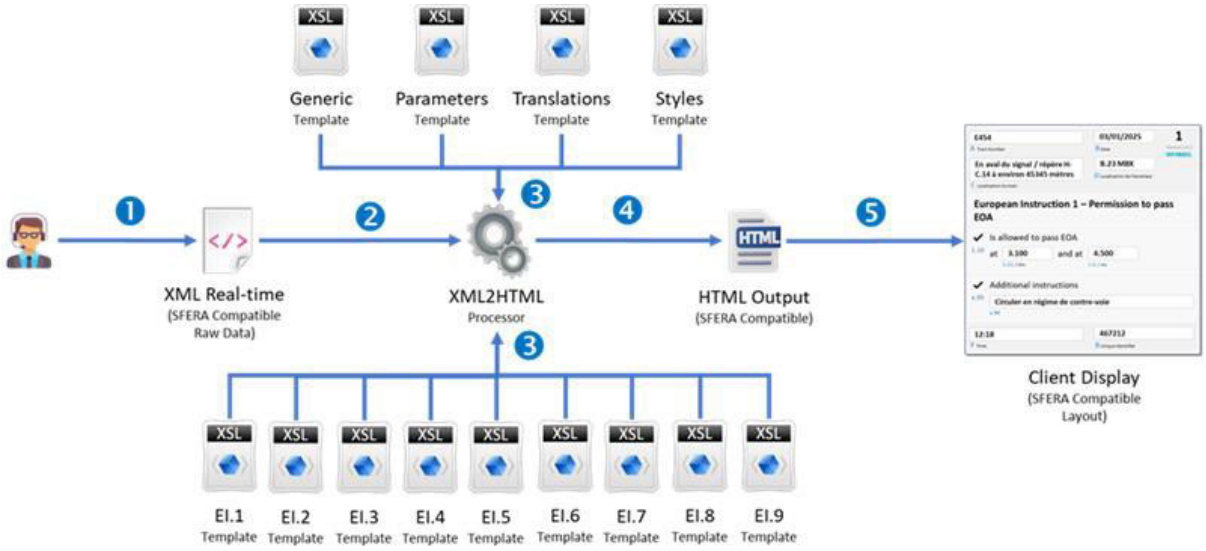


Fig. 16 - SFERA XSLT Software Architecture

1. The signaller inputs the instruction into their application, which exports it in a SFERA-compatible XML format in real time.

2. The real-time XML is received by the target device.

3. The XML2HTML processor on the target device combines the real-time XML data with the XSLT stylesheets and templates for the transformation.

4. The XML2HTML processor produces a SFERA compatible HTML output.

5. The HTML is rendered in the client application.

Appendix I - page 91 contains sample XSLT files that may be used as a basis for a workable solution to transform SFERA XML into HTML documents.

6 – Communication mechanisms

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

This point provides an overview of the SFERA communication protocol and authorisation mechanisms. Implementation specifications and details are described in Appendix B - page 84.

6.1 – Introduction

To accommodate the required flexibility in the communications and to match the separation of responsibilities between IM and RU, the protocol is designed as a set of layers. Each layer covers one or more features of the communications protocol and minimal technology choices have been made for each layer to provide interoperability.

SFERA specifies three main layers of the protocol:

- Data exchange
- Authorisation
- Communication

As illustrated in Fig. 17 - page 72, the data exchange layer is systematically based on a common SFERA XML message format. The authorisation and communication layers used may vary depending on the setup:

- TAF/TAP Common Interface between the IM SFERA-TS and RU SFERA-TS link
- Specific protocol for the two links involving the SFERA-OB

Fig. 17 provides an overview of this choice of architecture.

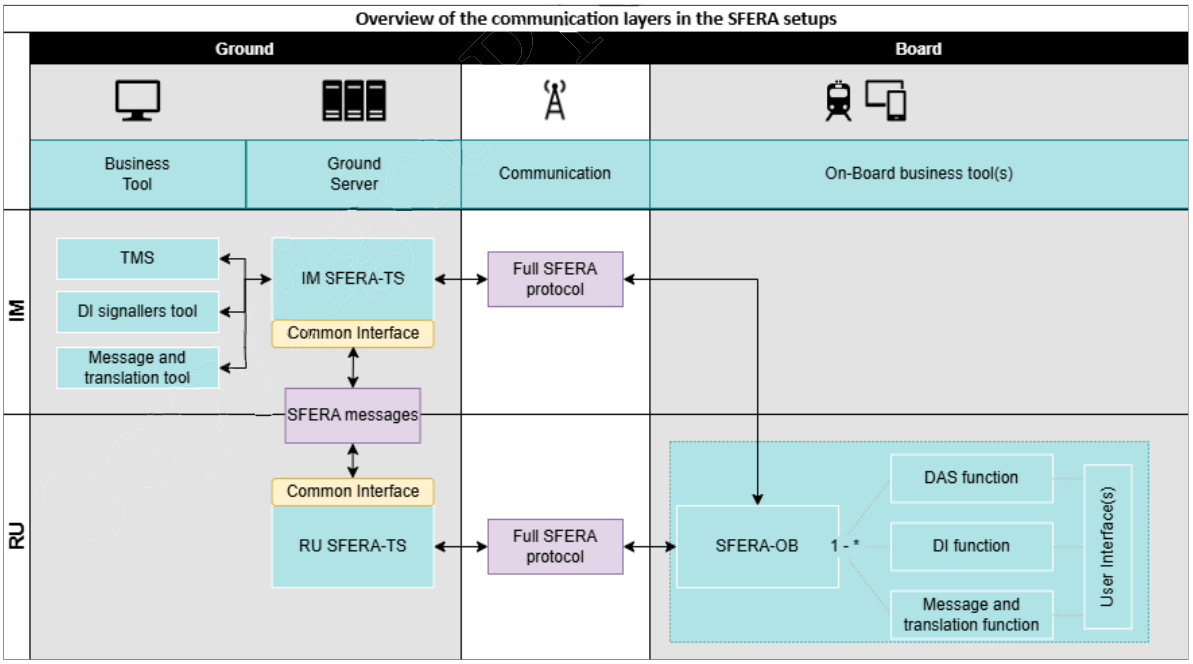


Fig. 17 - Overview of the communication layers in the SFERA setups

Fig. 18 - page 73 provides a functional decomposition which describes the interaction and communication between ground and on-board systems in the IM-RU setup. In addition to exchanging data (XML), which forms the core of this IRS, there is a layer which handles the communication between ground and on-board systems.

Depending on the link, this communication layer defines a bi-directional message exchange mechanism.

In the RU SFERA-TS <-> IM SFERA-TS link, the communication and authorisation layers are handled via the Common Interfaces deployed by the IM and RU.

For links involving the SFERA-OB, a communication layer has been defined to ensure guaranteed delivery over public Internet connections. Therefore, there is an authorisation layer which handles authorisation between trackside and on-board units.

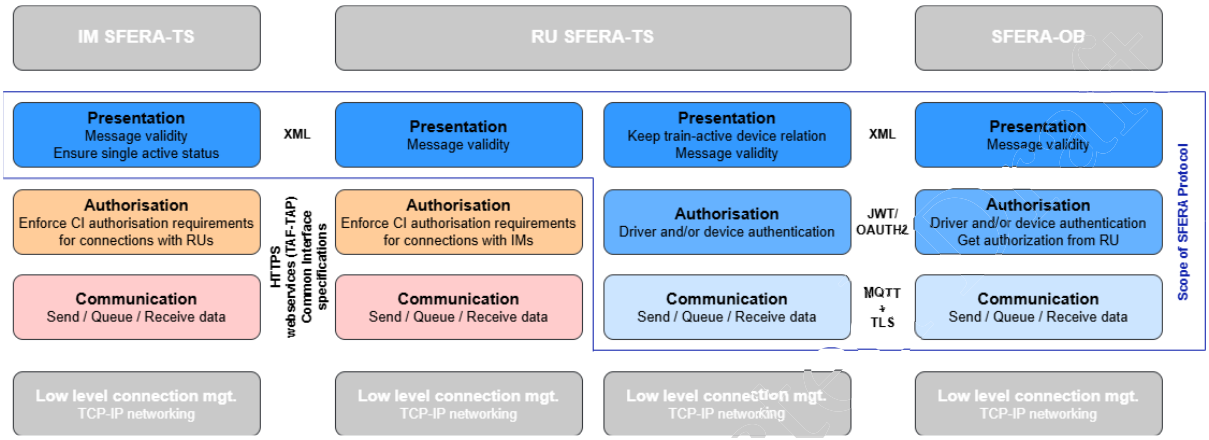


Fig. 18 - Different layers within the scope of IRS 90940 for the IM-RU Setup

In the IM-RU setup, the implementation layers are covered by two different specifications:

- The blue layers fall within the scope of this IRS and defined in detail in Appendix B - page 84 for the communication and authorisation layers.

- The orange layers fall within the scope of the *TAF/TAP TSI* specification of the Common Interface.

Appendix B provides an overview of the Common Interface protocol and details on how the SFERA XML messages shall be used to ensure full compatibility. The full specification of the Common Interface protocol is available in the *TAF/TAP TSI*.

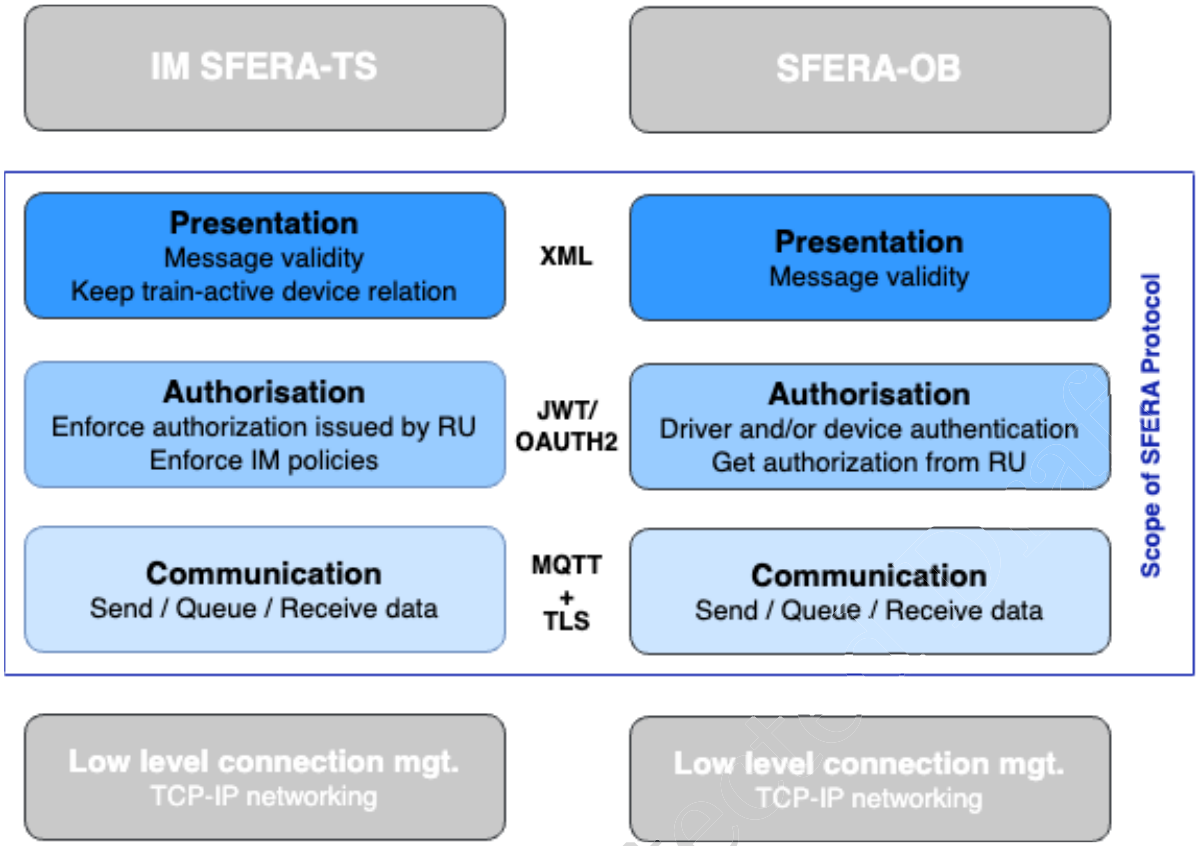


Fig. 19 - Different layers within the scope of IRS 90940 for the IM-Train Setup

In the IM-Train setup, the single implemented link specification is covered by the blue layers within the scope of the *IRS 90940*. These are also the same for the SFERA-OB/RU SFERA-TS link of the IM-RU setup. This commonality facilitates the implementation for RUs running trains across multiple IM networks with differing setups.

This IRS does not define all behaviours in detail. In certain cases (e.g. the driver/device authentication process), the implementation decisions are left to the parties involved.

6.2 – From use cases to data exchange

The SFERA protocol at the application layer is used across the three links defined in this IRS. Based on the use cases taken to build the SFERA protocol (as described in point 3.1.4 - page 48 and Appendix A - page 83), four basic communication patterns between the SFERA-TS and SFERA-OB (either directly or through the RU) can be identified:

1. SFERA-OB requests information from the SFERA-TS, which sends a reply with the requested information (e.g. request for a JP or SP), the received information is then confirmed with a reply message from the SFERA-OB to SFERA-TS.

2. SFERA-OB sends an information update to the SFERA-TS to inform it of a change in situation (e.g. a position update), which is acknowledged by the SFERA-TS.

3. SFERA-TS sends an update message to a train's SFERA-OB to inform it of a system change (e.g. a JP update), which in some cases requires a confirmation reply message from the SFERA-OB.

4. SFERA-TS requests information about a train to its SFERA-OB, which sends a reply with the requested information (e.g. a status update).

All high-level use cases are implemented through sequences of one or more of these communication patterns. Each pattern is realised as a set of interactions with the messaging system to transmit the data.

6.3 – Layered protocol

The SFERA protocol consists of three layers to logically and flexibly organise its features (see Fig. 17 - page 72). A proper SFERA implementation (on the OB, RU, and IM sides) should respect this structure.

6.3.1 – Presentation layer

The presentation layer handles XML data and ensures consistency in data exchange with a TMS, independently of the transport mechanisms addressed in lower layers. The key components to this layer are:

- XML validation and logical data validation to prevent corrupt data from being processed by or sent from the signaller's tool. This includes strict conventional XML schema validation, as well as additional logical value checks. Invalid messages received from the board devices should generate error replies

- Minor version checks and optional version conversions, if needed for the TMS

- Ensuring that only messages received from the active on-board device are forwarded to the signaller's tool

- Managing the switching of the active on-board device from one unit to another.

6.3.2 – Authorisation layer

6.3.2.1 – SFERA protocol

The SFERA communication mechanism enforces a strict separation between IM and RU responsibilities:

- The RU owns the on-board devices and is responsible for its drivers, the process of allocating drivers to trains and for ensuring that the intended driver is operating the intended train. An RU is also responsible for device management and security.

- The IM manages trains (not personnel) and is responsible for delivering the necessary information to trains in an appropriate timeframe. It must ensure that information is distributed according to privacy laws and other regulations concerning different RUs operating on its network.

This separation of responsibilities is reflected in the authorisation layer. The basic principle is that the RU is responsible for verifying the integrity and intended use of a driver/device and for issuing a JWT token that ensures that the user/device is authorised to access the SFERA service. The IM grants access to the SFERA-TS based on a trust relationship with the RU.

The RU shall define a process to verify:

- The integrity of the device running the on-board business tool (depending on whether it is a built-in or handheld device)
- That the user is authorised to use the on-board business tool
- That the user is designated to act on the given train (e.g. as a driver)

The IM shall define a process to validate that:

- A received token is issued by a trusted RU
- The journey parameters (e.g. train number) for requested and submitted information match the issuing RU

In an IM-Train setup, the IM may choose to assume RU responsibilities for managing devices, driver accounts and the authorisation process, for example, to make a deployment more cost efficient.

SFERA does not define the user/device authentication procedure itself.

Refer to Appendix B - page 84 for detailed specifications.

6.3.2.2 – Common Interface

The Common Interface, used for the RU SFERA-TS <-> IM SFERA-TS link, includes a security layer that is defined in the *Annex D.2_, Appendix_ E* of the *TAF TSI* and is used for standard CI implementations. It is based on SSL/TLS Certificates.

6.3.3 – Communication layer

To achieve interoperability in operations, standardisation at the technical level of data exchange is essential. The communication layer ensures reliable communication and manages the logic for distributing messages to the right on-board devices, effectively decoupling the signaller's tool from the on-board devices.

6.3.3.1 – SFERA protocol

The technology selected for implementing the communication layer in the SFERA protocol is Message Queuing Telemetry Transport (MQTT) Version 5 over a secured WebSocket (TLS). MQTT is specified in *ISO/IEC 20922:2016*, WebSockets are specified in *RFC 6455* and TLS in *RFC 8446*.

The rationale for this choice is explained in Appendix B. The use of *MQTT* applies to the following links:

- The IM-Train setup
- The RU SFERA-TS <-> SFERA-OB link of the IM-RU setup

6.3.3.2 – Common Interface

The Common Interface, used for the RU SFERA-TS <-> IM SFERA-TS link, implements a communication layer defined in *Annex D.2_, Appendix_ E* of the *TAF TSI*, based on HTTPS web services.

6.4 – Architecture overview

Fig. 20 illustrates the components of the DAS RU/IM ecosystem and the different setups.

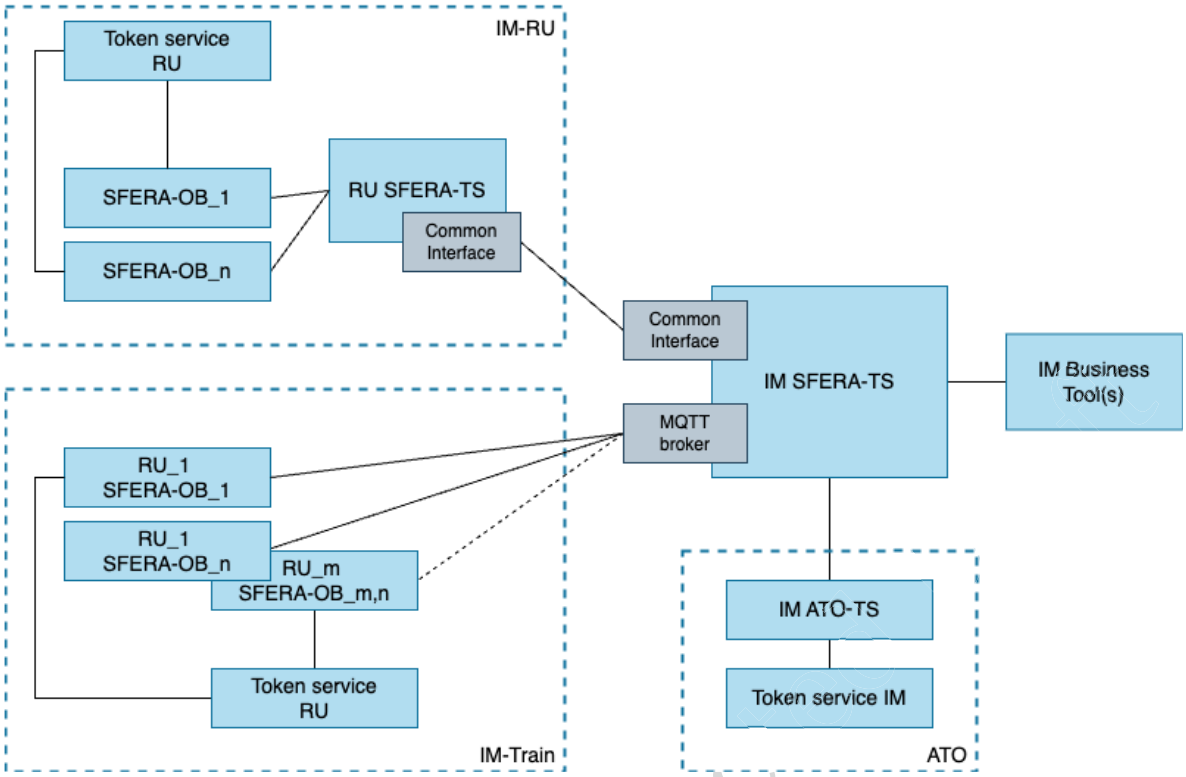


Fig. 20 - DAS RU/IM ecosystem

This IRS does not specify how the TMS integrates with the MQTT server. For other data exchanges, the design specifications given in Table 12 apply.

Table 12: Design specifications for different setups

Setup	**Design specification**
IM-Train setup	Data exchange between SFERA-OB and IM SFERA-TS **shall** be implemented according to this IRS (all layers).
IM-RU setup	For the IM SFERA-TS <-> RU SFERA-TS link, the application layer **shall** be compliant with the SFERA XSD, and the authorisation and communication layers **shall** comply with the Common Interface specifications. The exchanges between the SFERA-OB and RU SFERA-TS are the full responsibility of the RU. However, it is encouraged to implement the complete SFERA protocol (all layers) to facilitate transitions between IMs that may provide only one setup.
ATO setup	Data exchange between the IM ATO-TS and IM-SFERA-TS/DAS **should** be implemented according to this IRS (all layers). A fully proprietary solution is permitted.

7 – Maintenance of SFERA

See Reference documents - page 7 and Abbreviations and Symbols - page 10.

7.1 – The need for maintenance

There are several internal and external reasons for periodically updating the SFERA protocol. IMs or RUs have needs that cannot be fulfilled by the current version of SFERA; new legislations may require adaptations, technological developments or threats (e.g. cybersecurity) may require changes, and future international standards (such as TAF/TAP, IFC, EULYNX, RTM) or new versions of ERTMS/ATO SUBSET-126 are published. A procedure must therefore be in place to address these evolving requirements.

7.2 – Procedure to update SFERA

SFERA can be updated via the following processes:

- **Change requests**: Change requests may be submitted to UIC by mail or via the dedicated webpage, specifying the desired changes and providing a contact person for further consultation. Details to submit are given on the UIC website.

- **First evaluation**: The SFERA secretariat (i.e. UIC-mandated to coordinate SFERA activities) will conduct an initial evaluation to establish whether a change request requires a faster response time from the SFERA working group than the regular half-yearly cycle. If so, the secretariat will consult the SFERA working group chairs and convene a meeting.

- **Periodic update**: The SFERA working group meets periodically to process all received change requests. The SFERA secretariat will prepare the invitation together with the chairs.

- **Small and large changes**: The update procedure aims to balance speed and thoroughness. To serve SFERA users effectively, the SFERA working group is authorised to publish new versions of SFERA without consulting all UIC members for changes with limited impact.
  For changes with significant impact on the structure and/or functioning of SFERA, a formal UIC publishing procedure will be followed. The working group consists of experts and is therefore empowered to make this distinction.

- **Publication**: All versions of SFERA will be published on the UIC website along with all supported version and release notes explaining the differences from previous versions. These notes will include a migration strategy. Minor changes (backwards-compatible with earlier minor versions) may take effect immediately.

- **Questions and support**: UIC members or companies wishing to build solutions using SFERA, can reach the SFERA project management a SFERA@uic.org .

- **Complaints**: If a UIC member disagrees with how a change request was handled or denied, the SFERA secretariat may be contacted for clarification.

7.3 – Making solutions compatible with the latest version of SFERA

Compatibility with SFERA can be ensured as follows:

- **The need for upgrading solutions**: One of SFERA's goals is to facilitate interoperability. If an international train operates in three countries where each IM supports a different major version of SFERA, this goal is undermined. Therefore, for each implementation, IMs and RUs (and their supporting manufacturers) should establish procedures to manage upgrades and maintain compatibility with the latest version.

- **Minor and major updates**: SFERA makes a distinction between minor updates and major updates:

  - A minor update is a non-breaking change to the data structure AND semantics.
  - A major update is a breaking change, either to the data structure, semantics, both data structure and semantics, or the messaging and authentication mechanisms.

  Version numbers will reflect this, with minor upgrades changing from e.g. Version 2.3 to 2.4, and major upgrades from e.g. Version 3.2 to 4.0.

- **Establishing procedures for upgrading**: To maintain compatibility while upgrading, IMs and RUs should mutually agree on the procedure to be followed. Here, it is strongly recommended that solutions are able to support two major versions for a limited period of time. The following rules support effective lifecycle management during this process, with a fair division of responsibilities between trackside and on-board:

1. SFERA major and minor version numbers are included in every exchanged message.

2. The SFERA-OB should support all minor versions within the same major version (e.g. by ignoring unknown elements and attributes, and new attributes/elements being qualified as optional).

3. The SFERA-TS should accept all minor versions within the supported major version.

4. If data conversion between versions is required, this should be carried out by the SFERA-TS.

5. The SFERA-TS must support previous major version(s) for an agreed period.

6. The SFERA-TS may upgrade to a new minor version at any time.

7. The SFERA-OB shall upgrade to a new major version within the agreed migration period.

These rules assume that SFERA-TSs are typically centralised systems with sufficient resources and control over software rollouts. In contrast, the SFERA-OB may consist of many different types of devices from different owners and manufacturers, often with limited resources and less control over deployment timing (due to the number of devices).

Breaking changes in the underlying messaging system (MQTT) or JWT token specifications are very rare. If they occur, they will require separate service endpoints at the SFERA-TS, and the migration strategies should be considered on a case-by-case basis. Therefore, most SFERA version updates can be implemented without changes to the underlying messaging system. Major changes are more likely to involve modifications which make the XSD incompatible with older versions (e.g. making optional elements mandatory, or changing enumerated values).

7.4 – Identifying opportunities for enhancement

The SFERA project maintains a list of NSPs (Network Specific Parameters) implemented in different SFERA implementations. The shared file contains the technical identifiers defined by each SFERA provider and the signification.

The SFERA Technical Subgroups will review the updates of this list to identify the opportunities for harmonisation and enhancement of the structured parts of the SFERA XML messages. The main candidates would be NSPs implemented by different SFERA providers which refer to the same notion, but unique NSPs could also be considered.

7.5 – Visibility of the deployment of SFERA

Companies that are implementing SFERA are invited to share the main characteristics of their SFERA Trackside through a shared file hosted by the SFERA Working Group.

This file can be used for different purposes by members of the SFERA community, for example:

- Identifying cross-border implementation opportunities
- Establishing a way to contact project teams
- Business model for suppliers.

Appendix A: SFERA Use Cases

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix B: SFERA Communication Protocol

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix C: SFERA implementation cases

The SFERA implementation cases illustrate the way the different layers of the SFERA protocol interact in different operational scenarios. The focus is on the communication layer and the use cases, taking external events that may impact operational use into account. They also describe the driving mode that the SFERA-OB, the SFERA-TS and the signaller's tool will read.

The main objective of introducing implementation cases into the SFERA Group workflow was to assemble the individual protocol layers into a comprehensive overview. This helps identify inconsistencies between different aspects of the protocol.

Implementation cases are included in the IRS as they may be useful:

- For those seeking to understand how the SFERA protocol functions,

- As a basis for discussions between RUs and IMs when considering implementation options for a given network.

The implementation cases are described solely from the IM-Train setup perspective, as this setup encompasses all layers of SFERA. In the IM-RU setup, the functional elements should be the same from both a device and IM SFERA-TS (mainly the driving modes). The RU SFERA-TS would only add a layer which summarises the driving modes of the different devices to transmit a unique status to the IM SFERA-TS (the driving mode of the active device, if any).

All the implementation cases can be realised using the SFERA protocol. However, there is no obligation to implement all of them on a given network or for a given RU. Furthermore, the list of implementation cases is not exhaustive. The scope of implementation should be agreed upon between individual RUs and IMs.

A full list and description of the implementation cases is provided in an external appendix to this IRS, available in the "Attachments" tab of the *UIC IRS 90940* publication page.

Appendix D: SFERA XML Schema Definition

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix E: SFERA Data Handbook

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix F: Sample SFERA XML Communications

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix G: Correspondence Table SUBSET-126/ SFERA

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix H: Error codes in SFERA

This appendix contains a list of error codes for SFERA messages. The list is provided as an external appendix (via the *UIC IRS 90940* publication page), as it may need to be regularly updated following requirements changing during the SFERA maintenance process (see point 7 - page 80).

Appendix I: XSLT samples for DIs

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix J: Supporting documents

This is an external appendix to this IRS, available via the *UIC IRS 90940* publication page.

Appendix UIC-1 - Regulatory listing

This appendix is an integral part of the following IRS:

IRS-90940:2026, Ed. 3: SFERA Protocol: Train Operation Data Exchange with On-Board Devices

The table(s) below list(s) the regulatory source(s) specifically citing this IRS of which UIC has been made aware by the relevant bodies. If applicable, users shall check the validity of this table(s) with the source indicated.

Table UIC 1-1: European regulatory sources citing *IRS-90940:2026, Ed. 3*

Source: title	Source: passage	IRS: passage targeted	Remarks
none			

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Bibliography

The following documents are recommended for the understanding or for more in-depth information about subjects treated in this IRS:

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[5] *Wikipedia: Punktförmige Zugbeeinflussung.* Available: https://de.wikipedia.org/wiki/Punktf%C3%B6rmige_Zugbeeinflussung

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